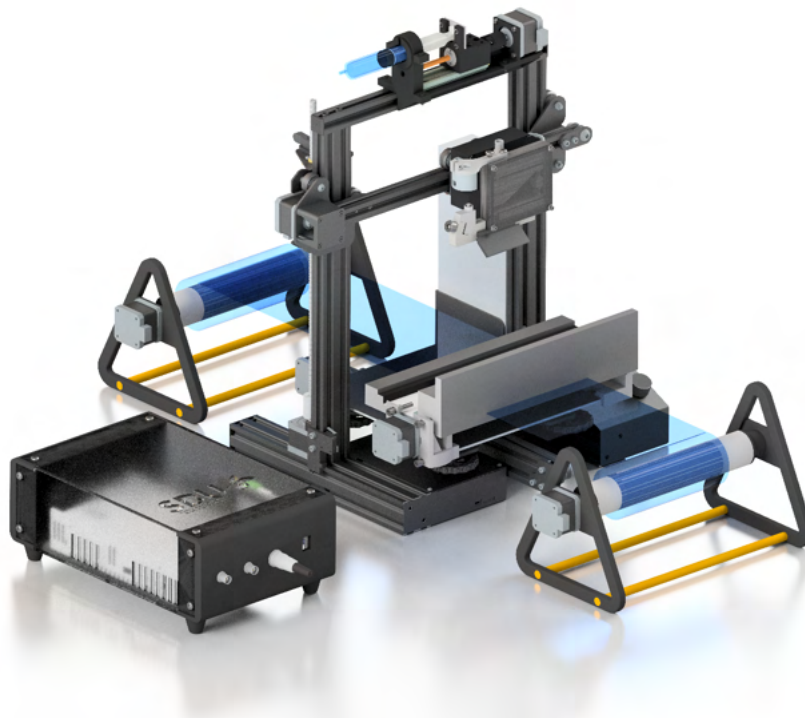


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Bachelor Project

R2R Spotter Plotter



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Abstract

This bachelor project details the development of a low-cost, scalable, non-contact micro-dispensing system, termed the "R2R Spotter Plotter," designed as an add-on unit for standard 3D printers. The primary objective was to enable the generalized and scalable deposition of multiple chemicals in various patterns onto flexible foil substrates, specifically for the creation of chemical optodes.

The project involved the design and integration of several key subsystems: a piezoelectric jet-valve for non-contact dispensing, custom driver electronics, a roll-to-roll (R2R) platform for continuous substrate feeding, a syringe pump for precise fluid delivery, and a comprehensive control box. A bespoke desktop application was developed to provide a user-friendly interface, enabling grid definition, real-time spotting process visualization, and computer vision-based droplet footprint analysis.

Key achievements include the successful creation of a functional R2R platform and a robust, intuitive software interface that eliminates the need for manual SD card transfers. The jet-valve was designed with a quick clamp-on mechanism, allowing for easy integration with existing 3D printers. While the system demonstrated continuous spotting capabilities and met the ease-of-use and cost-effectiveness criteria for the add-on unit itself, it faced challenges in consistently achieving the targeted droplet footprint diameter of 300 microns, with repeatable droplets settling at approximately 650 microns. This limitation, along with issues related to fluid path sealing and constant pressure supply, prevented the full realization of sub-mm spacing and large-area grid patterning as initially envisioned.

Despite these performance shortfalls, largely attributed to project planning oversights regarding fluid dynamics, the project successfully demonstrated the feasibility of adapting a 3D printer for advanced micro-dispensing applications. It highlights the complexities of fusing mechanical, electrical, and software systems in a mechatronic context and provides a solid foundation for future improvements, including advanced parametrization, closed-loop control for droplet sizing, and the integration of multiple dispensing heads.

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1 Introduction

1.1 The R2R Spotter Plotter

The professional market for non-contact micro-dispensing systems is widely saturated with e.g. medical picoplotters like Scienion's sciFLEXARRAYER reaching down to 10 picoliters per droplet for precise multiplex assays within diagnostics. Equally industrial jetting valves like Nordson's PICO Pulse reach down to 500 pico-litres per droplet jetting thick adhesives like glue, silicon or solder paste. Unfortunately, those and similar systems are expensive and often come as bulk, single purpose space-consuming units making the technology unavailable for small labs or hobbyists.

Their base construction does not differ from a 3D printer and nowadays every lab has at least one 3D printer or can get one extremely cheap. The idea of this project is to develop an add-on unit that can easily be hooked up on a 3D printer transforming it fluid dispensing unit. Mounting an sophisticated piezoelectric jet-valve like the [V2000](#) onto a 3D printer would showcase a possible solution for upscaled spotting but its price rules out the simple add-on requirement. For many lab-based engineering experiments, requirements are much less strict and there is potential for a solution that can be adapted as a DIY 3D printer add-on. Therefore, a new technology can be introduced into the lab without committing to a new machine requiring large funding.

In addition, standard pico spotters are limited to individual substrates. We want to enable scalability with an additional add-on unit, that will allow rolls of flexible foil substrates to be patterned with solution grids in a roll-to-roll fashion (R2R).

In previous work [\[3\]](#) the need for a multi-chemical nanoliter dispensing unit was addressed by a 3D printer modified to a contact dispenser. This came with limitations in maximum optode sizing, hence the overall aim of this project is adapting a technical solution for the scalable and generalized deposition of multiple chemicals in various patterns onto the same substrate at sub-millimeter spacing utilizing a 3D printer.

2 Problem formulation

2.1 Background and Rationale

During my last student assistant position at SDU-Nanosyd I was tasked with developing a fluid dispensing solution for the automation of in-house binder deposition onto wafer substrates for the [PRECISE](#) project and successfully built a platform for precise fluid dispensing by modifying a 3D printer with a syringe head and software [Figure \(1\)a](#) and [Figure \(1\)b](#).

During the development, Klaus Koren's research group at the department of microbiology at Aarhus University got in touch regarding finding a technical solution for the creation of a new optode by spotting polymer solutions in distinct patterns (mm size) on PET foils for chemical optode detection ([Figure \(1\): c, d & e](#)). [Figure \(1\)-d](#) shows the foils being used under water (with sand) in normal and under UV light. [Figure \(1\)-e](#) shows extrapolated data from the dyes, calibrated to indicate oxygen levels, spatially distributed across the water-sand interface.

The software, which initially was a python script, was then adapted with a graphical user interface and scalability options for different dot-grids. Each grid's settings could then be tuned to fit the specific viscosity of the different chemicals. Due to the size of the first iteration vacuum chuck the spotting was limited to an area of 20x40mm ([Figure \(1\): a](#)), with room for upscaling only limited by 3D printers print bed sizes. The advantage of using this low volume spotting method within optode-sensors is that the following analysis can show the distributional behavior of different chemicals inside the same space at the same time.

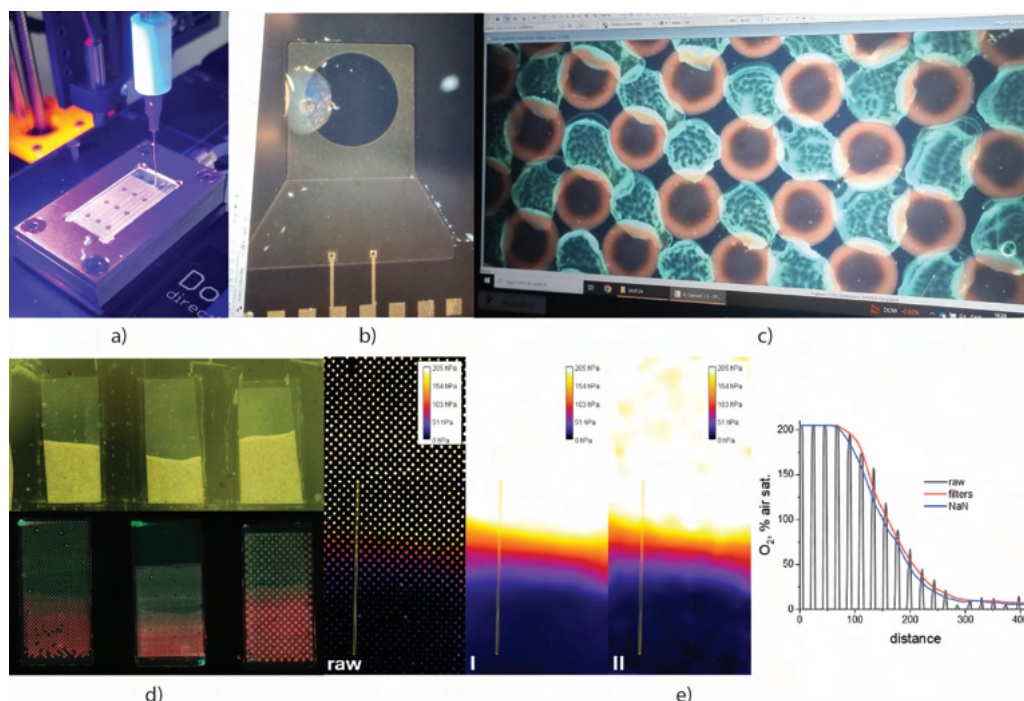


Figure 1: a) Spotting process, b) cantilever spot (slight miss), c) 10x magnification, d) sediment test under uv light, e) O2 Sensor analysis

Building on this work a new scientific paper was then written in collaboration of SDU and Aarhus University, which as of this moment is submitted for review to the Advanced Optical Materials journal and made available as a preprint at: [Modular Grid Optodes for Simultaneous Multiparameter Imaging via RGB Detection: a Spatially Patterned Approach to High-Resolution Chemical Sensing](#).

As this concept has been proven on small-scale, the real interest lies in implementing this at a large scale, to fit more devices and enable larger area monitoring e.g. for continuous analysis of soil until meter-depth, monitoring whole root systems etc. The latter has been previously shown with single-chemical planar optodes within the MARTINIS project [8]. Consequently, creating new optodes for that machine would provide a great basis for further research.

2.2 Objectives

The objective of this project is to find a solution to the scalable creation of said optodes, that comes at a low-cost entry point and with focus on being able to integrate with the existing systems in the lab, e.g 3D printers or motion platforms.

To transform the previous project towards scalability it is important to move from contact to non-contact dispensing. The reason for this is, that contact dispensing requires a repetitive positional accuracy in height of 100 to 500 microns relative to the substrate [3]. Although achievable on flat surfaces with extensive calibration, this technology would risk needle crashing or lack of droplet contact when using flexible substrates, as planned for the R2R. Additionally, contact dispensing introduces a significant time constraint as the needle needs to move up and down for every dispensed droplet.

Therefore the main goals of this project are to:

1. Design a dispensing unit for non-contact jetting, that does not require aspiration in order to spot grids in a continuous manner. The performance requirements are set aiming to achieve the precision

of the grids created by the previous contact spotter.

- (a) Which therefore are:
 - Re-achieve the previous spotting footprint of 300 micron diameter, e.g 2.9 nL (7)
 - Implement continuous spotting to fill two A4 pages in a 600-micron grid, requiring a volume of 1 mL (8)
 - Aim for 100 Hz spotting (9)
 - (b) Calculate the requirements for driver electronics to validate a commercial solution and design the driver if none can be found.
 - (c) Implement the control signals for the driver electronics to work with or around Marlin.
 - (d) Keep the unit price below €250 (ENDER 3 PRO value)
2. Create a software and application so users without programming skills can use the dispensing.
 - (a) Live visualization of parameter inputs
 3. Design, develop and install an R2R platform that will enable printing on larger sheets. The platform should also be an “add-on” to a 3D printer.
 - (a) Reuse Y-axis motor for R2R driving (possible belt switching)
 - (b) Upgrade Marlin firmware for infinite Y axis movement
 4. Redesign the unit to have a mechanism that is user friendly and able to quickly “hook-up” to a 3D printer.
 - (a) Clamp-on mechanism (remove need for FDM print head removal)
 5. Add a multiple solution dispensing unit.
 6. Implement remote control of the system through the application (remove need for SD-card.)

Success criteria

The simple add-on requirement is considered successful if the final product can be implemented for less cost than the 3D printer it is used on. Additionally, the firmware changes cannot reduce/remove the functionality of FDM printing while the add-on is not in use and the software will need to be manageable by users capable of performing basic slicing tasks for 3D printing.

The clamp-on mechanism is considered successful if mounting and unmounting of the jetting module takes less than 5 seconds. The design of the dispensing unit is found successful, if the performance requirements as in Figure (1)a) will be achieved. Finally, having a simultaneous multiple solution dispensing unit is considered a “nice-to-have” and not critical to project success.

The aim of the project is not to compare the performance of the device to industry leading Jet-Valves or the like, in terms of accuracy, minimum volume and general sophistication of the design. This project will evaluate, what is possible to achieve, without precision components and will show, what kind of devices laboratories can reproduce at the respective lower price points.

2.3 Project plan

Upon starting this project, 4 phases had been defined and each estimated to take one month. During the project ideation the focus lays on reading literature, analyzing what technology is used on the market

and finding, how a solution could be implemented in a DIY fashion. The following design and component choosing part of the ideation phase would then seamlessly transition to the mechanical design phase, as parts become available and solutions would be found. All components and sub-systems should be finished by the end of the second month to allow time for programming and testing and validation. This would leave space for additional requirements which were expected to be found only during the progression of the project, e.g. the need for a camera vision to create data measuring the performance of the system.

This can be illustrated by a Gantt chart, reporting on the project's progress at the end of the second month and providing a respective estimation of future tasks:

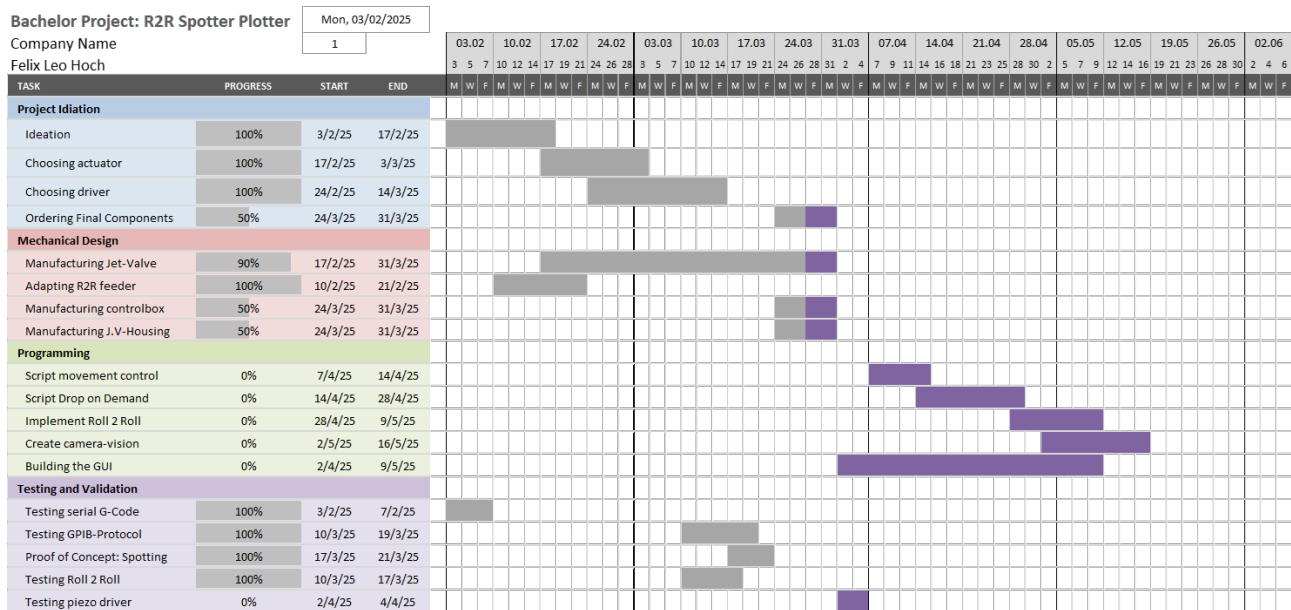


Figure 2: Gantt chart for the project after progression of 2 months

2.4 Risk assessment

The project has three major sub-goals that are independent of each other. The failure of one sub-goal does not necessarily lead to the failure of the project. The sub-goals are:

- **Jet-valve:** The jet-valve is the most interesting part of the project. If it will not be possible to build a working jet-valve in-house, it could either be bought from a commercial supplier or the project could continue with the contact spotting head from the previous project. The risk of jet-valve failure leading to project failure is low.
- **R2R-feeder:** The Roll 2 Roll system is a mechanism that is well established in the industry. Because industrial components are widely available and it is possible to reuse old components like paper-feeders from ink-jet printers or similar, the risk of failure is low. A failure of the R2R-System would not lead to total project failure, but would limit the maximum scalability to the size of the 3D printer's build plate.
- **Software:** As the software is based on sending custom G-Code commands to a modified 3D printer mainboard running a custom Marlin flavor, there are several possible failure points.
 - Marlin could be too low level to support real-time G-Code serial communication. The Jet-Valve needs real-time position feedback for true Drop-on-Demand operation. Without this, the system would be slow, inducing non-scalability and project failure.

- The project could require a too large codebase to be implemented in time.
- Implementation of high level firmware like Klipper could be out of scope for the project.

Testing of the system could fail at several points.

- The system could not be able to handle the strong solvents THF and Toluene.
- The system could not be able to create a working droplet and that one at either required speed, size or pattern.
- Testing could not be finished due to running out of time.

Concluding, the project is expected to be a spot-on task for a mechatronics engineer. Designing a system like this requires subsystems of mechanical engineering, knowledge and analysis of electronic components and the ability to program a custom software solution fusing them all together. The understanding of the system as a whole with a view into each subsystem, is what the field of mechatronics is all about.

3 Literature review

A start to implementing a non-contact dispenser as a 3D printer add-on is to look towards the industry and how such a task is handled in that environment. As previously stated, the market is widely saturated and can be categorized into two major groups. On the one side is the field of life sciences, which uses non-contact dispensers for miniaturization of analytical assays. Within this field samples of e.g. cells, enzymes or antibodies are precious and expensive. Utilizing jet dispensers reduces waste, ensures high repeatability and minimizes risks of cross contamination due to the automated process. This enables the technology to have especially high leverage for e.g. drug discovery, where the sheer amount of possible combinations to mix elements is simply unmanageable, if done by hand [21]. The great interest in this field can be put to scale by the fact that for example Scienion's products alone have been used in 1725 publications within life sciences. Their solutions contain piezo-dispense capillaries (PDC) creating dispensing pressures through piezoceramic tube actuators for the range of 220 to 800 pL and electro magnetic micro-valves for handling single droplets from 25 nL to 1 μL . Another approach based on directional pipe squeezing of flexible tubes was presented by Hamilton [24] and offers piezostack driven jetting ranging from 1 to 70 nL with the advantage of having disposable tips and fluid paths, eliminating the risk of cross-contamination.

On the other side there are industrial Jet-Valves with specializations for glue dispensing for electronic packaging, general manufacturing solder paste applications or even automated greasing for bearings. There are numerous companies like Nordson EFD, Delo, Y&D Technology, Axxon, Techcon and many more, who all present needle stroke based products capable of jetting liquids from low viscosities to thick solder pastes or hot melt glues. Their impact for the automation manufacturing can also be shown, by the great interest for optimization and parameterization of the technology as shown by Li et. al [5].

Thanks to this research about needle jetting is widely available, e.g. Huang et. al [25] presented nozzle and needle combinations, Lu et. al [20] simulated droplet formation, Wang et. al [6] identified jetting defects like satellite droplets and many more have contributed already to any concern that might come up.

Comparing those products and looking at the price range it becomes evident, why using one of the professional valves for this project is out of scope and budget. After talking to industry partners it also became clear that for their customers the cost of the system is not of importance. They value precision, reliability and possible downtime, hence they buy the system that comes as close to 100% runtime as possible.

Company & Name	Technology Type	min Volume	Continuous Frequency	Price
Scienion sciFLEXARRAYER S3	Piezotube	200pL	NA	45000€+
Hamilton PipeJET Kit	Piezostack	1nL	340Hz	7720€
DELO-DOT D4	Piezostack	2nL	200Hz	16721€
DELO-DOT PN3	Pneumatic	2nl	140Hz	11390€
Y&D Technology V2000	Piezostack	1nL	1400Hz	4286\$
Techcon TS9800	Piezostack	0.5nL	1500Hz	4900€

Table 1: Technical comparison of dispensing technologies from selected companies.

Since the project expects to work with high viscosity polymer cocktails, needle jetting is further investigated.

3.1 Basics of jetting operation

The basics of jetting can therefore be condensed to a needle accelerating the working liquid in the direction of the nozzle and which if accelerated enough, breaks the surface tension and capillary forces at the nozzle and ejects from the nozzle(Figure (3) b)).Next, the needle closes the flow by touching the nozzle, the flow breaks off, and a flying droplet is formed(Figure (3) c)), upon which the process is repeated. Changing the drive characteristics of the needle, adjusting stroke distance, velocity and waveform therefore has a direct impact on the created droplet volume, shape and whether unwanted satellite droplets will be created.

Illustrated by Figure (3):

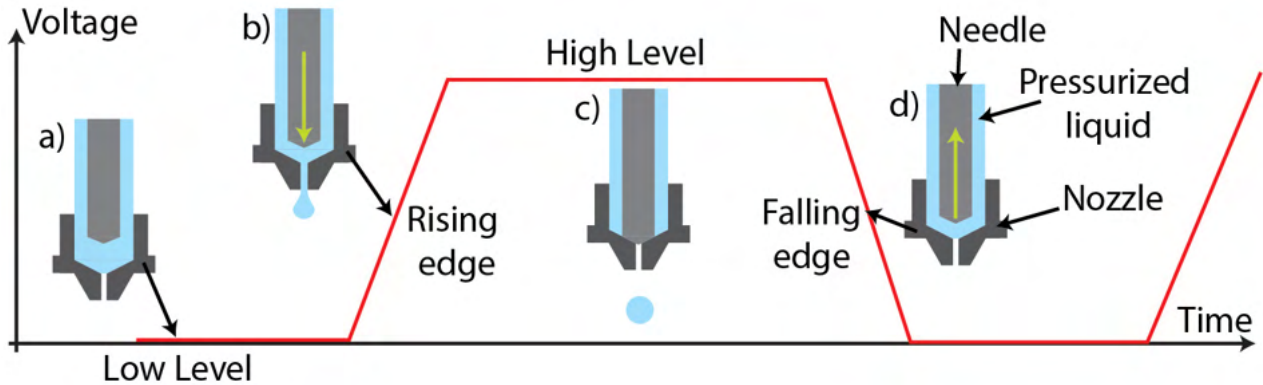


Figure 3: Basic droplet formation

The droplet volume increases with increased stroke and pressure, while decreasing at increase in drive frequency[27][6][9]. Satellite droplets are created by overflow droplets at high velocity[6]. Therefore, the drive voltage should be reduced and/or the rise time should be increased, if satellite droplets are encountered during testing. As Trimzi[10] confirmed the minimum droplet of a system can be created if open time of the nozzle is reduced and frequency increased. Another finding of importance by Trimzi is, that with increase in frequency, and thus a reduction in opening time, the difference of droplet mass becomes less dependent on the nozzle diameter, making it possible to create minimum diameter droplets both with, for example both 0.2 mm and 0.4mm nozzles.

4 Choosing a dispensing mechanism

Initially the following systems had been given consideration:

- car petrol injector rebuild,
- pressure spray,
- solenoid actuator jet ,
- piezo spotting,
- voice coil actuator jet,
- pneumatic jet valve.

An interesting concept would have been to rebuild a fuel injector, as the internal components are exactly the same, as in a Jet-Valve system. The only difference is, that they are set up in the opposite way, using pressurized fuel as the acceleration method and the needle as a valve. Taking a fuel injector apart and putting the actuator back in reverse could have been a low budget solution, but had been discarded after open source jet-valve mechanisms were found. The pressure spray was also discarded as it suits applications where surface coverage is the focus rather than creating single spots.

As the Jet-Valve was chosen as the non-contact dispensing solution, the next consideration was on how to drive the needle. An intuitive thought when requiring an oscillating motion is to use a voice coil actuator. Similar to a speaker, those actuators can change their stroke and oscillation based on their input amplitude and frequency. Due to their large travel range and inverse non precise steps in the $100\mu m$ range they could not been used. As SHI et. al [27] described, the needle stroke is required to be precisely controllable between 0 to $300\mu m$. This also excluded solenoid actuators from the selection, as even short range proportional solenoids do not offer satisfactory resolution.

This left only piezoceramic actuators and pneumatic systems, which are also those used in the industry. Because industrial Jet-Valves driven by piezo technology are rated to deliver higher performance and accuracy, even when suppliers compare to their own pneumatic [systems](#), this project will try to build a Jet-Valve based on the piezoceramic actuators.

4.1 Open-source mechanisms

During the project ideation phase three different open source mechanisms had been found and given consideration. The single lever hinge mechanism presented by Trimzi et. al [10], the single long lever mechanism by Wu et. al [9] and the double amplification mechanism by SHI et. al [27].

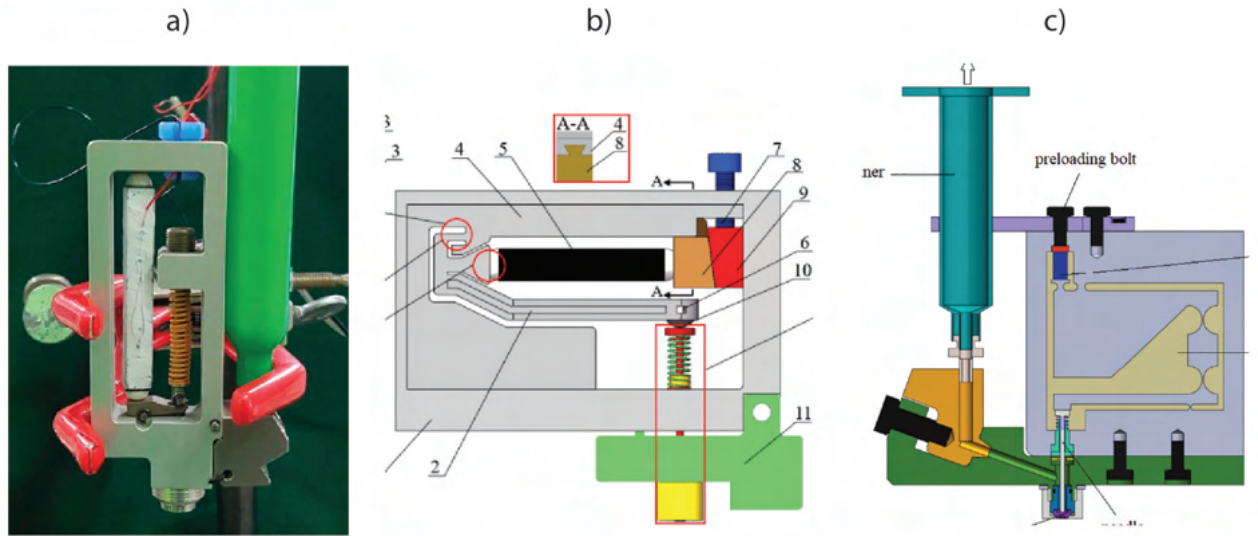


Figure 4: Jet-Valve mechanisms by: a) Trimzi et. al [10], b) Wu et. al [9] and c) SHI et. al [27]

As piezoelectric actuators scale linearly in price and total displacement, mechanism c) of Figure(4) was chosen, as the cheapest piezoceramic actuator to be found with suiting properties would cost 63€, leaving room for the development of the rest of the valve and staying within the 250€ budget for the valve body.

4.2 Choosing driver electronics

In order to control the stroke of a piezoelectric actuator a driver circuit is required. Typical benchtop amplifiers are capable of driving a wide variety of different actuators and can easily be tuned to fit specific experimental use cases. Such a driver could be implemented here as well but is out of scope for this bachelor project, as it is expensive and its use outside the project is uncertain. Some examples:

Driver Name	Voltage Range	Frequency Range	RMS Current (μ F)	Price	Link
PiezoDrive PD200	0–200 VDC	0–230 kHz	1.2 A	2139 \$	[14]
Thorlabs BPA100	-30–150 VDC	20 Hz–100 kHz	2.5 A	2161,29 €	[23]
Physik Intrumente E-619	-30-130 VDC	0-20 kHz	5 A	3-4000 €	[12]

Table 2: Comparison of different benchtop piezo drivers

Because the scope of this project includes a singular use case, with specific design properties, a commercial off-the-shelf (COTS) solution can be used, if found with suitable electrical properties. Therefore, the following analysis will go into depth about how to find the required electrical properties. If no COTS can be found, the analysis will form the basis building a custom amplifier circuit.

4.2.1 Design requirements

Using the top-down approach, all properties of the driver can be found by looking at the actuator and the desired maximum frequency.

- Piezoelectric Actuator:

Dimensions: 5x5x10.5mm

Capacitance: $4.2 \mu F \pm 15\%$

Resonant frequency: $115 kHz$

Drive: sin, pulse, sawtooth or square wave

- Operation:

Maximum required frequency: $100Hz$

Duty cycle: variable

4.2.2 Equivalent Circuit

In order to accurately represent the piezo during simulations an equivalent circuit model has to be used. Several commercial Piezo Drivers such as the modules PDu100B V3 [16] and PDu150 [17] as well as the bench-top amplifiers like PD200 [14] model the piezo actuators as simple capacitors, which also fits, as the manufactures of the piezo only state its capacitance, impedance at resonant frequency and dissipation factor.

According to the IEEE Standard on piezoelectricity [1] the equivalent circuit can be represented by the Van Dyke Model near the resonant frequency and as a simple capacitor for frequencies higher and lower. Newer models can take the real part of the impedance at lower frequencies into account and are compared by Liu et. al [26], but as the projects application requires simple low frequency charging the Van Dyke Model for non-resonant frequencies will be sufficient. Charge and discharge times will therefore be dependent on the series resistance of the driver and capacitance of the piezo.

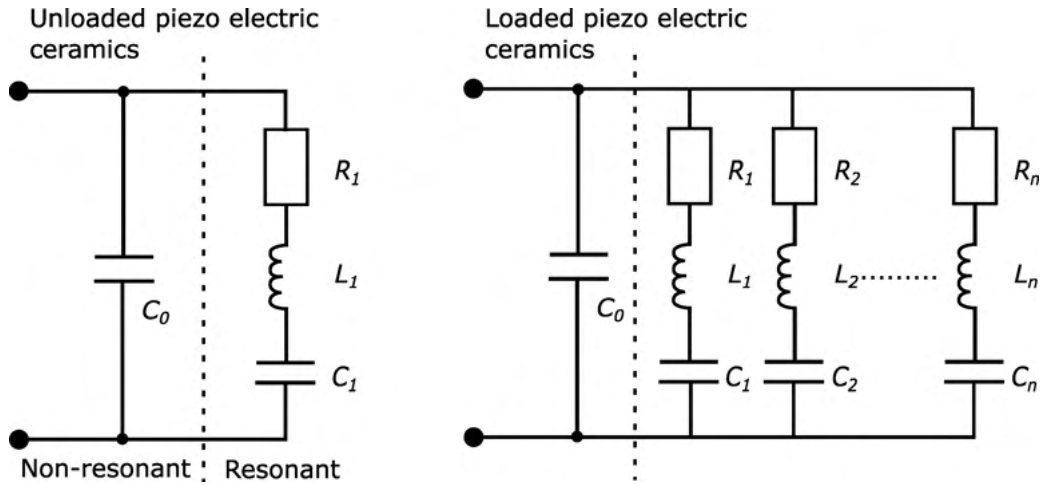


Figure 5: Van Dyke model of piezoelectric elements

Due to the mounting of the piezo actuator inside the amplification mechanism, its state changes to a loaded piezoelectric ceramic, which introduces all the resonant modes of the touching mechanisms into the equivalent circuits. As the modal analysis for the amplification mechanism found its first mode to be at $436Hz$ [27] and the required frequency is $100Hz$, the equivalent circuit can be modeled by a single capacitor. This is, because the resonant nodes of the Van Dyke model only add to the equivalent circuit if it is driven at the resonant frequency, which is not recommended for this application.

4.2.3 Design of the Drive Stage

The next step is choosing a method of driving a capacitive load. Previous work suggest a totem pole drive stage for microrobotic applications [2], while other evaluate different circuits, also concluding that push-pull/source-sink configurations [4] are a necessity for low power drivers. Other integrated circuits like the DRN2700 [22] utilize differential amplifiers for sourcing and an additional MOSFET for sinking. Multiple other driving circuits have been proposed by Sam Robinson [19] which also utilize variations of a push-pull follower.

Because multiple established drivers use a complementary symmetry amplifier, a simplified version will also be used here. The required minimum current supply can be determined by fitting the rise time in an RC circuit to the maximum desired frequency. To account for material variations, a positive tolerance of 15% and an additional 20% fluctuation buffer are included to ensure the peak voltage is always reached.

An RC circuit consists of a resistor (R) and a capacitor (C) in series, where the characteristic time constant (τ) of the circuit is given by:

$$\tau = RC$$

Given that the system reaches approximately steady-state after 5τ and the maximum drive frequency is $100Hz$, $\frac{T}{2}$ sets the maximum rise time for a 50% duty-cycle square-wave.

$$\frac{T}{2} \approx 5\tau$$

Substituting $\tau = RC$, we get:

$$\frac{T}{2} \approx 5RC$$

Solving for R :

$$R = \frac{T}{2 * 5C}$$

Substituting $C = 4.2\mu F * 115\% * 120\% = 5.796\mu F$ and $T = \frac{1}{f} = \frac{1}{100Hz} = 0.01s$

$$R = \frac{0.01s}{2 * 5 * 5.796\mu F} = 172.5\Omega$$

The maximum inrush current is then found by Ohm's law

$$I = \frac{V}{R} = \frac{75V}{172.5\Omega} = 435mA$$

For which the LTspice simulation confirms a rise time below $\frac{T}{2} = 0.005s$ and the maximum current.

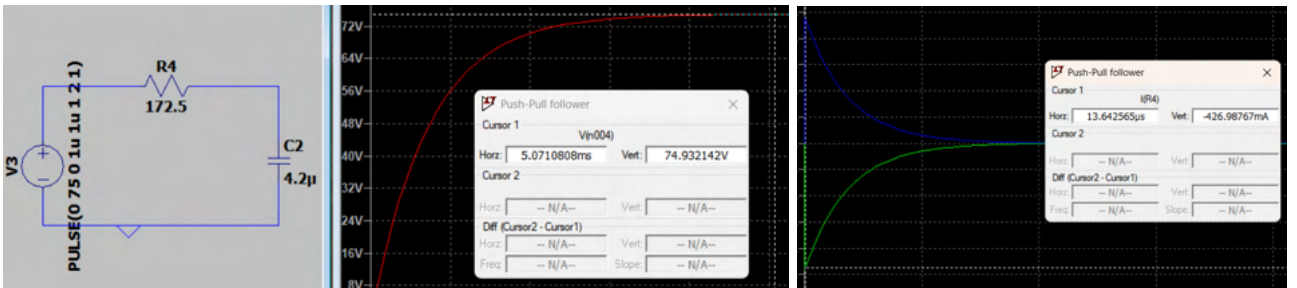


Figure 6: First order RC-circuit

A more interesting performance metric is the RMS current, as it describes, what power the driver actually has to deliver while switching at the desired frequency. For this, a drive stage was simulated in LTspice. The circuit design is based on a push-pull follower as previously defined and components are selected to be able to drive the piezo actuator with a maximum voltage of 75V.

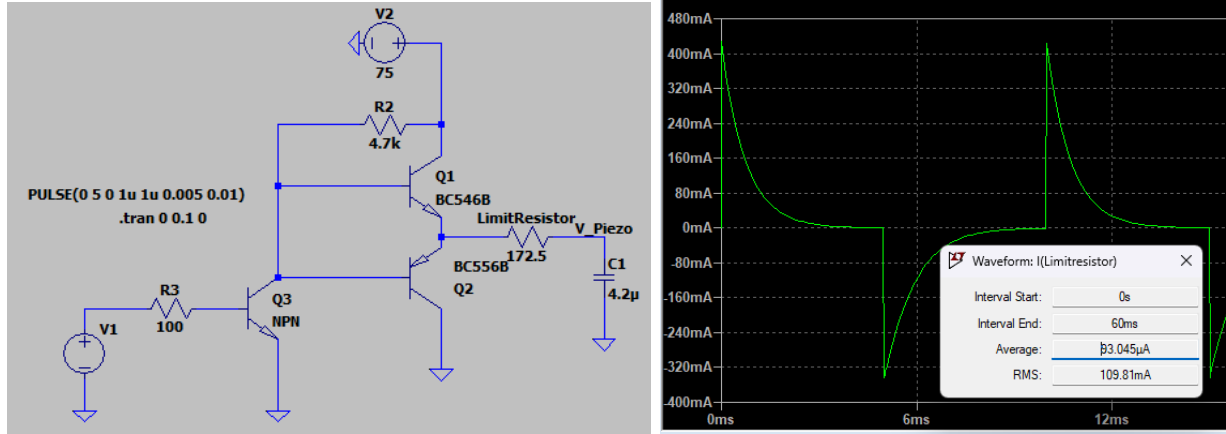


Figure 7: Push-pull driver circuit

The simulation (Figure (7)) then provides the RMS current to be 110mA.

Equally important is the slew rate of the driver, which describes how fast the output voltage can change and is used by manufacturers to describe performance. Given:

$$\Delta V = 75 \text{ V}, \quad \Delta t = 5 \text{ ms} = 0.005 \text{ s}$$

The slew rate (SR) can be calculated as:

$$SR = \frac{75 \text{ V}}{0.005 \text{ s}} = 15,000 \text{ V/s}$$

Convert to volts per microsecond:

$$SR = 15 \text{ V}/\mu\text{s}$$

Therefore, a COTS driver would need to be capable of a slew rate of at least $15 \text{ V}/\mu\text{s}$, while providing a minimum RMS current of 110 mA and sustaining a peak current of 435mA, to drive the piezo at required dynamics of 100 Hz at 75V.

Driver Name	Voltage Range	Slew Rate	RMS Current	Peak Current	Price	Link
PiezoDrive MX200	0–200 VDC	$60 \text{ V}/\mu\text{s}$	550 mA	1 A	700 €	[13]
Physik Instrumente E-610	-30–130 VDC	/	60 mA	140 mA	920 €	[11]
PiezoDrive PDm200	0–200 VDC	$20 \text{ V}/\mu\text{s}$	120 mA	300 mA	320 \$	[15]
PiezoDrive PDu150cl	-30–150 VDC	$38 \text{ V}/\mu\text{s}$	235 mA	300 mA	401 \$	[17]

Table 3: Listing of suitable COTS driver

Finding the MX200 driver provides a suitable solution for the project which leaves headroom, for any unexpected current requirements induced by the amplification mechanism. After several consulting talks,

the MX200 was also recommended for the application by SI Scientific, the supplier of PiezoDrive drivers in Germany. As their quote came at 700€ for the driver, the budget had to be considered.

If the budget did not allow for the MX200 to be purchased, the project requirements could be adjusted to fit either the PDm200 or the PDu150cl. As the only missing fitting metric was the peak current, this could be solved by reducing the maximum required spotting frequency.

If those would also have been out of budget, the project could be continued by building a custom driver circuit, based on the previously mentioned push-pull driver. This would require a further deep dive into power electronics. Since the the focus of the project was on system design and integration, the additional funding for the MX200 was sourced by the project supervisors.

To drive the MX200 a controllable voltage source from 0-10V is required, hence a Keysight 33220A function generator was reserved for use in this project. The function generator is then going to be the interface between driver electronics and software, as it is programmable.

5 Creating the R2R Spotter Plotter

5.1 Jet-Valve design

The creation of the Jet-Valve used in this project is based on several steps. First, the chosen design had to be remodelled and validated by another simulation in CAD. Then it can be combined with a housing to be a robust solution with wiring and finally combined with the 3D printer base by a snap on mechanism.

5.1.1 Adapting the original design in CAD

This project reproduces the displacement-amplifying mechanism proposed by Yihong Shi and Fu [27]. Because not all design-parameters of the valve had been stated in the paper, the remaining ones have been manually adjusted and then validated by fitting to the simulation results of the reference paper.

From the paper [27] we obtained the parameters corresponding to Figure (8):

Candidate point 1	Value	Length/Position Parameters	Value
R1 (mm)	1.3464	L1 (mm)	2.3
R2 (mm)	5	L2 (mm)	44.8
R3 (mm)	1	L3 (mm)	5
b1 (mm)	1.9417	L4 (mm)	11
b2 (mm)	2.5	L5 (mm)	3.5
b3 (mm)	0.4	L6 (mm)	10.4
b4 (mm)	0.35	X1 (mm)	5.7
Total deformation max (mm)	0.244	X2 (mm)	20

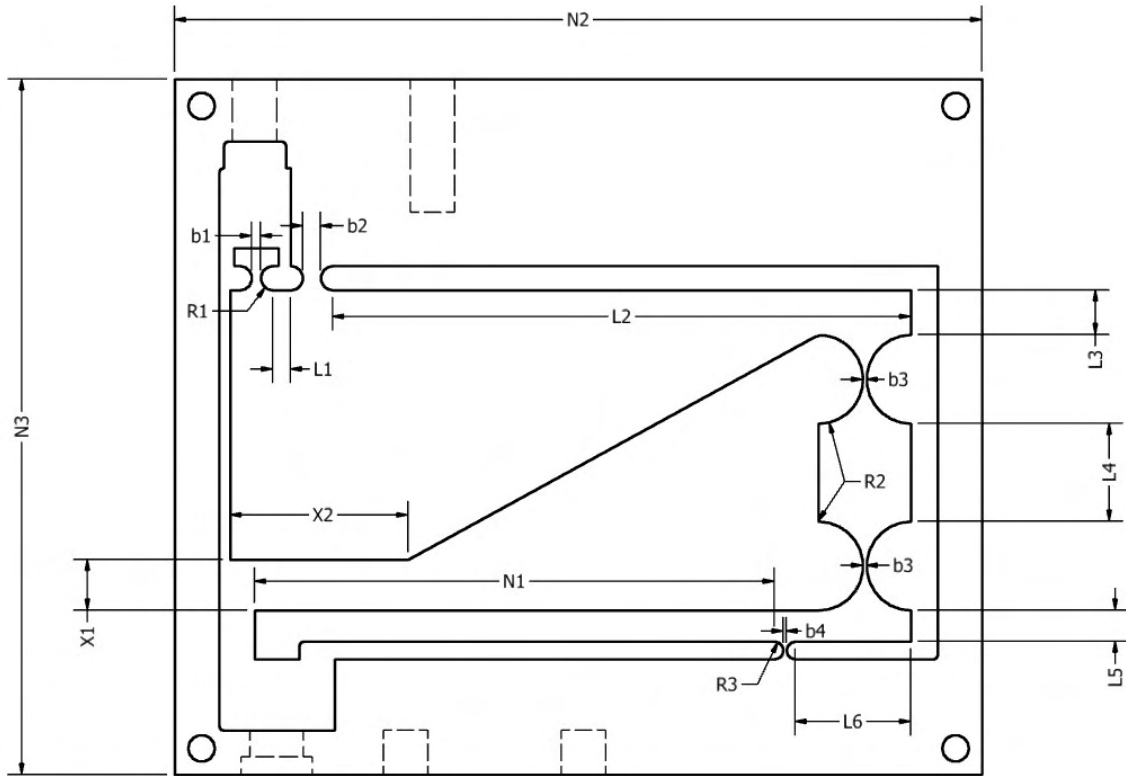


Figure 8: given parameters with valve drawing

For which N1, N2 and N3 had been undefined and could be chosen to fit the assembly. At this point a simulation of the design was created for validation, that the required output displacement was actually achieved. The simulation (Figure (9)) was setup in Autodesk Inventor Nastran, with the material properties for spring steel stated by SHI (density: 7820 kg/m³, elastic modulus: 197 GPa, Poisson's ratio: 0.282), a node element size of 1.4mm and a linear constraint, which moves the surface touching the piezo downwards by 11.2 μm , as this was the specification of the piezo bought from Thorlabs.

With the original design's parameters this resulted in a displacement of below 150 μm , which was not satisfactory to start manufacturing. The difference in results might come from different simulation softwares, as SHI uses ANSYS and the project had access to Autodesk Inventor Nastran. Another reason might be different constraint conditions, which had not been defined by SHI. To proceed, the design parameters were iteratively adjusted, starting with those having the highest sensitivity on the output displacement as described by SHI et. al [27]. After several iterations satisfactory results of 0.2356 μm were achieved by

elongating the lever and fine tuning the radius giving:

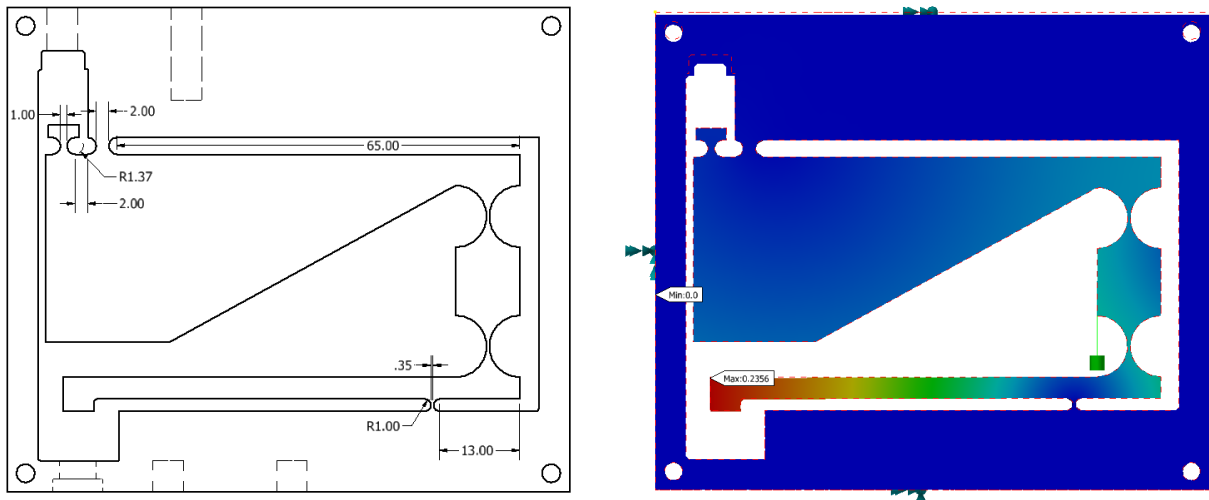


Figure 9: final parameters and simulation results

As the adaptation of the design parameters create a big change in the inertia of the amplifying mechanism, it is important to notice, that the first mode as referenced by SHI, resulting in resonance at 436Hz, will not be accurate anymore. Therefore, a new modal analysis was conducted and the results showed the new first mode at 249Hz. As the objectives aim for a maximum jetting frequency of 100Hz, this is still satisfactory.

Further design considerations

The fluid body plays the crucial role of combining the valve body, with a fluid inlet and the output nozzle. Initially, the original design was remodeled, and sub-components were identified. At the time of the design phase, suppliers for components like the nozzle, nozzle fitting and needle assembly had not been found, as SHI[27] did not share the complete component list of the project. Hence, the workshop at SDU was contacted and the contact with local manufactures was recommended. A request to SIM-TEC.APS for manufacturing a precision nozzle (50) with hole sizes down to 100 micron and the later calls had been ignored. As of next professional manufacturing service CAMCON-IT, who were conveniently located at the SDU Sonderborg campus had been contacted and responded with a recommendation for a japanese supplier [Union Tool Co](#) for drills with 20 to 100 microns diameter. Even though finding these drills was possible, running them would require a spindle at 100,000 rpm and laser measurement equipment, rendering the task impossible to do in-house. In addition the expected price of (startup: 400-700 DKK, material and hours: 500-1000 DKK) totaling between 1000 to 2000 DKK for a single nozzle resulted in exceeding budget and the cheap add-on requirement. Hence, another solution had to be found.

Nozzle choice

The thought at hand when already using a 3D printer as the base is, why not to use a 3D printer nozzle as well? Their high demand makes them cheap and widely available, while their threaded design makes them easy to replace and seal with teflon tape. The biggest concern with using such a nozzle is the unavailability of datasheets. Various suppliers do not give any data sheets at all and others like RS-PRO only list general dimensions, but excluding information about the hole depth and cone angle of the drill. Without this information it is required for the design to be adjusted for different hole depths.

This can be done by adjusting the pretensioning screw of the valve, or by simply using less turns of the thread on the nozzle. Hence, the unavailability of datasheets was selected as minor problem, and it was

decided to continue in this direction. As SHI showed [27], optimal cone angle of the needle is in parallel with the nozzles angle. Hence, the nozzles angle needed to be found. For this, an attempt to cut a standard nozzle in half on the lathe, failed. As standard nozzles also come with fluid inlets of 2mm and the chosen needle diameter was also 2mm, it was attempted to expand the nozzle inlet, which also failed to be done with precision.

At last an MK10 (37) steel nozzle was found with the satisfactory dimensions, featuring a 4mm fluid inlet and nozzle diameters from 200 to 400 microns available. In addition, a section view image is shared, from which a cone angle of 80° could be estimated, which appeared to be a perfect fit.

This adaptation of the design can be illustrated by Figure (10):

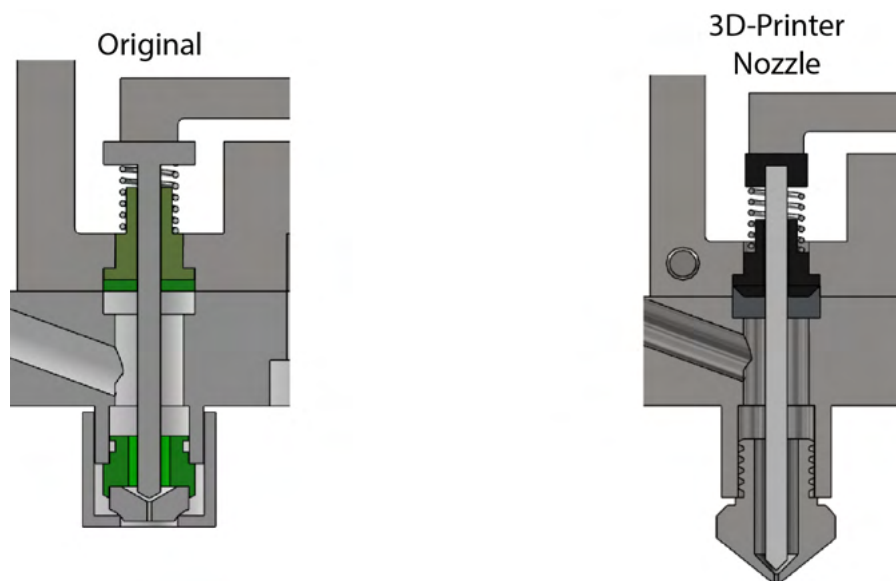


Figure 10: Original with turned steel components (green) and precision nozzle, And adapted version with 3D printed components (black) and 3D printer nozzle

Needle assembly

As a needle assembly supplier had initially not been found, a trial was made to create it in a DIY fashion. Industrial Jet-Valves are using needles, milled out of a single piece of steel, creating a solid connection between backplate and needle as seen in the original image of Figure (10). Because this part would also be expensive if custom made, an attempt was made to use a 2mm steel rod and glue a 3D printed backplate onto it with a cyanoacrylate adhesive (superglue). Conveniently the disassembly of the ink-jet printer provided a 2mm rod, that could be shortened to the required length and sharpened with a cone of approximately 120°. After the second iteration, this system proved to be robust and successful. Robust is defined here by not breaking while manual pulling and passing a function test of 150Hz, which corresponds to a safety factor of 1,5.

As there also had not been any information about the sealing used in SHI's Jet-Valve, an initial try was made to create a DIY teflon plate cut on the water jet. This later on proved to be un-suiting (8).

5.1.2 Manufacturing the valve

Having obtained all design parameters the valve body and fluid inlet were ready to be manufactured. For this, the local workshop at the SDU Sonderborg campus was contacted, while bringing a 3D printed version

of the parts, to explain what was the required task at hand and which tolerances did matter. After a few consulting talks, it was decided to try to cut out the shape with a water jet out of stainless steel. SHI presented the design out of spring steel, but it would be unavailable unless deciding to buy in metric tons, according to the workshop. Similarly SHI states to have used machining to manufacture the body, but due to the workshops concerns, that milling to wall thicknesses of 350 micron could introduce resonance in the part, leading to failure. Hence, water jet cutting was chosen. Water jetting also would not be the preferred technique for such thin walls because of the cone cutting nature in thick materials. As machines like wire EMDs were not available at side, which would deliver straight cuts, water jetting was tried, under the premise that in the case of failure, wall thickness and general parameters would require adjustments.

A successful cutout was made in the second iteration, since the first one was not cutting all the way through the material. The next step was then to add all the features by a subtractive manufacturing process. Not having worked with stainless steel parts before, made this part a challenge. With some failures like broken thread mills and some hand cuts, but after a learning process and the guidance from Jan and Jacob, the technicians in the workshop the parts had been manufactured to finish successfully, without requiring a second try.

Some parts of that process can be shown in sequence as Figure (11):

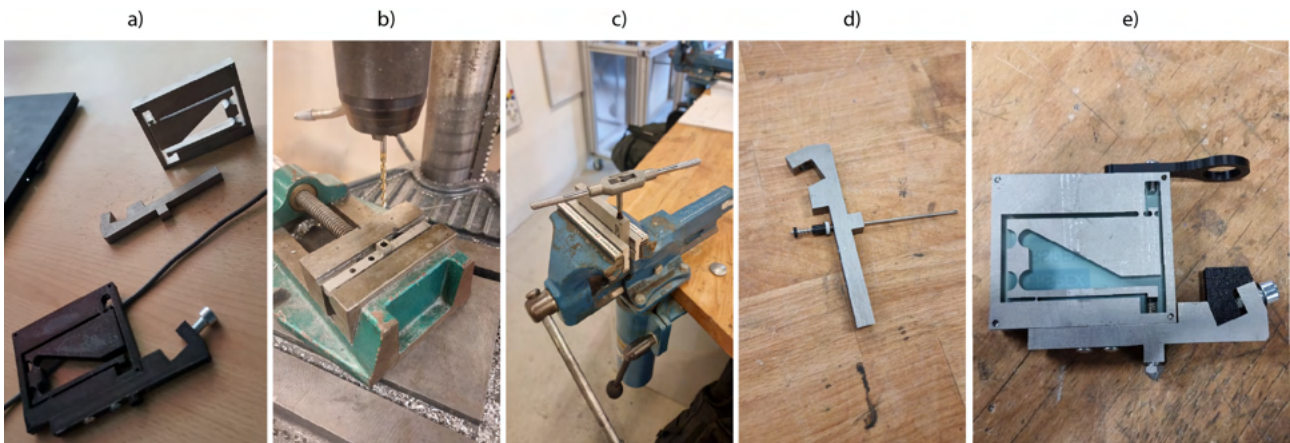


Figure 11: Manufacturing sequence of the valve: a) 3D printed master and water jet cut bodies, b) drilling of base holes for threads and inserts, c) creating threads, d) trial assembling needle assembly, e) finished base valve body

5.1.3 Converting to an add-on unit

For converting the valve to an AddOn unit, a housing was designed. This would then also include grooves where another part could snap into. The idea was then to have the snapping part permanently mounted onto the 3D printer. This part would then also handle the mount for a BNC plug, so no high voltage handling wires are ever exposed.

As this part was not presented in the reference paper, a new ideation had to be made. This was approached by first designing a backplate to the valve body, which includes space for the BNC plug and then adding triangular cutouts, where any snapping mechanism could be attach to. The bottom cutout was placed in the center, while the side cutouts were placed in the top segment, allowing for increased stability through the elongated lever and access to future hinges from the top of the valve.

Based on that, using a sketch (in Figure (12) & high resolution (40)) of the snapping mechanism was created to list all required design parameters. It was chosen, to make the bottom seat and the flexible hinges as screw-on parts to the baseplate, which was mounted to the printer. This allows the user to

adjust the tightness of the contact and simple exchanging of hinges without reprinting the whole part. In addition, the baseplate that mounts to the printer includes a mount for the BL-Touch probe, which is set to be levelled with the nozzle tip. The design sequence can be seen as Figure (12):

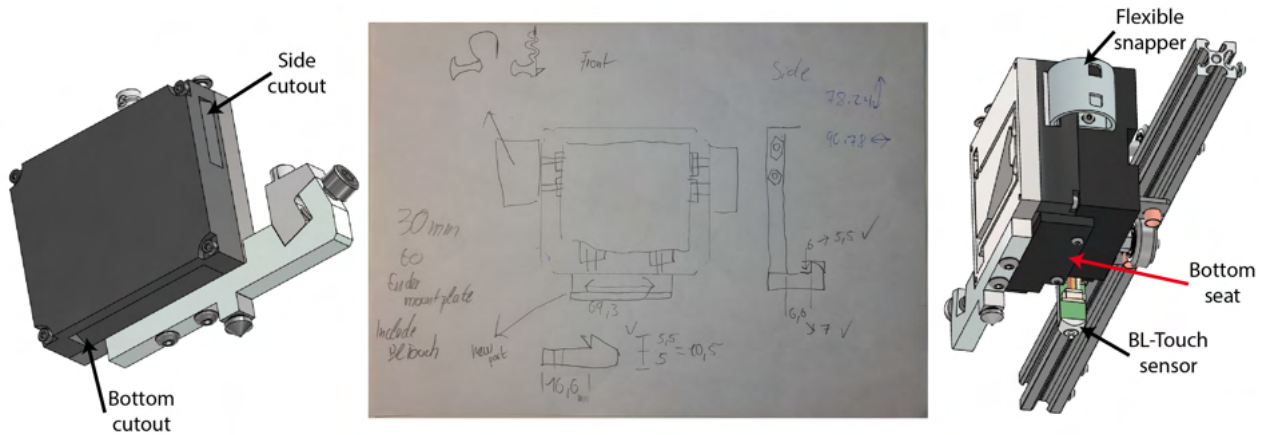
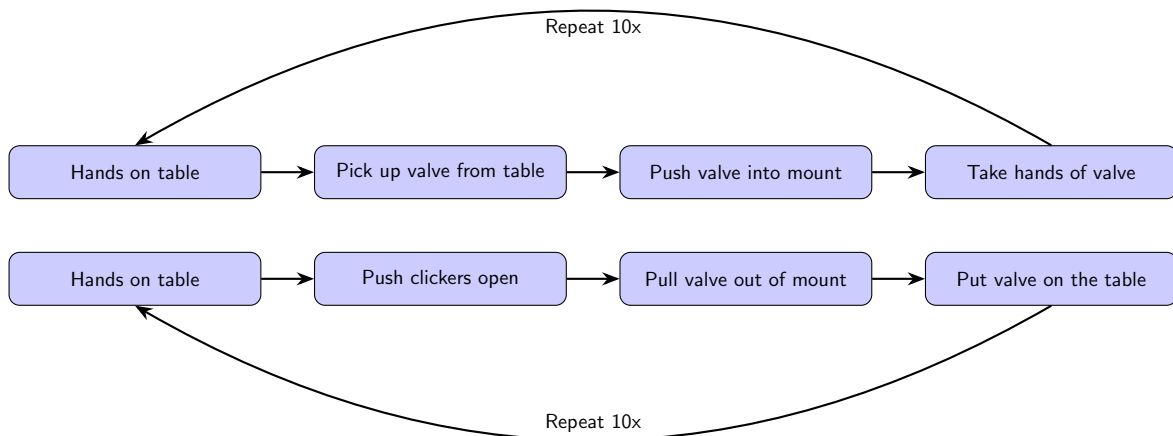
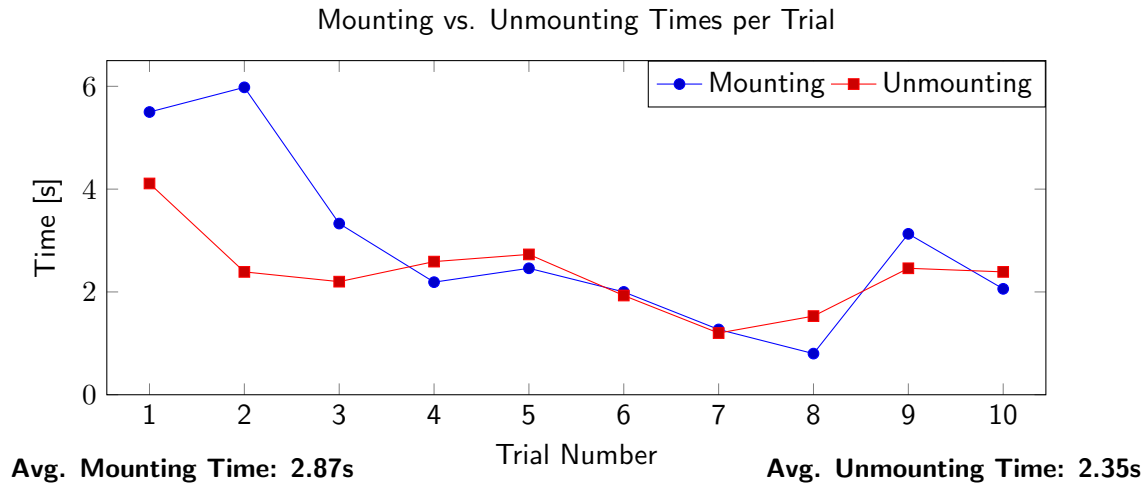


Figure 12: Design process of snap-on mechanism

To find a measure of performance, a small experimental study of the snap on mechanism was conducted. For that, the Jet-Valve was mounted and unmounted 10 times in a row and the time for each respective process was measured. The processes are as follows:





From the experiment (Figure (5.1.3)) it can be seen in Plot (5.1.3), that for both mounting and unmounting the average time is below 5 seconds, which satisfies the easy to use add-on requirement. The initial drop from the trials 1 and 2 to 4 and 5 is related to the operator getting used to the snap on system. The second drop from trial 5 to 8 is caused by the operator pushing the limits on how fast the task can be done. While at last, trials 9 and 10 represent a normal mounting and unmounting procedure of a trained operator.

The precision of the results has to be taken with a grain of salt, as the measurement includes the human error introduced by the reaction time of the second operator who was performing the timing. Nonetheless the results are more than satisfactory for this application.

5.1.4 Adding a camera

The last part of building the Jet-Valve was the mounting of the camera to the body. Because of its small and streamlined form factor, an endoscope camera was first mounted and tested on detecting spots. Due to the low quality of the camera's sensor and its unsuitable focus range of above 5cm, using it was not ideal. As a next step, it was found out, that even cheap desktop webcams can be modified to have a focus range below 10mm, creating great close-up imaging at 1920x1080 pixels. This was possible due to its modular lens, which at full extension allows for focus at 2mm physical distance.

Proceeding with the webcam camera, a mount was then made. It attaches the camera with an angle, that fixes the camera's focus point to intersect with the landing point of droplets, if the nozzle is positioned 10mm above the substrate.

The complete Jet-Valve Assembly is therefore seen in Figure (13):

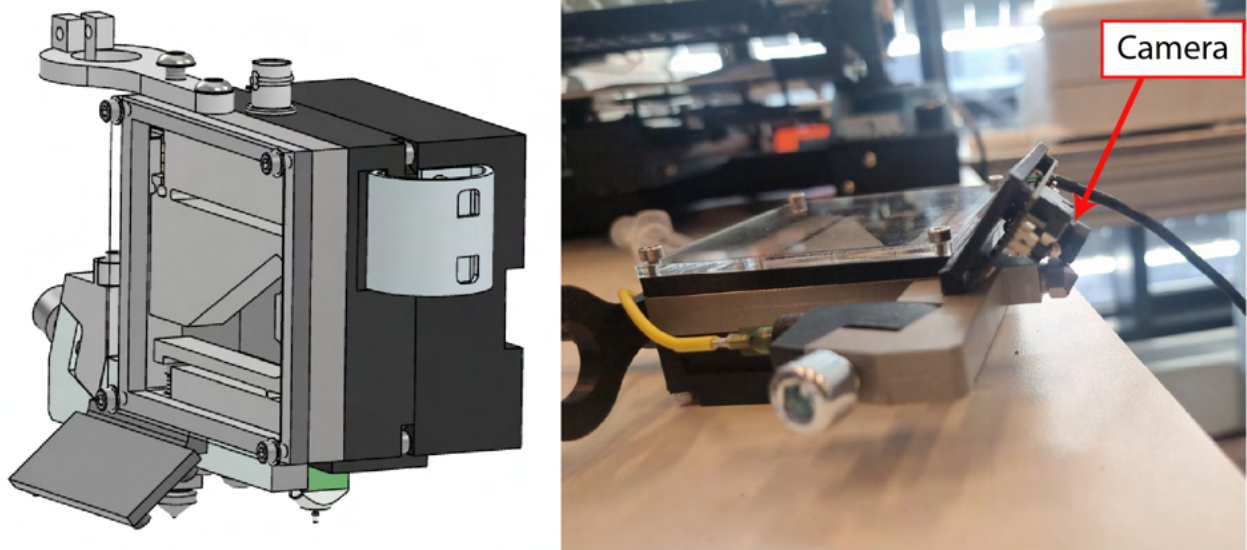


Figure 13: CAD and picture of complete Jet-Valve assembly

Technical information about the Jet-Valve can be found in the Appendix (13), (47), (48) and (49).

5.2 R2R mechanism adaptation

After looking at different approaches to R2R machines presented in the industry, it was clear, that those systems utilizing real conveyer belt technology would be an overkill in terms of supported load and speed. On the other hand, Neuralabel had an R2R inkjet printer in its catalog, the Calisto, that would perfectly fit the required task. Hence, the design of the R2R system was guided by the [Neuralabel GoPress](#).

Building a basic version of such a system requires:

- A paper feeder
- De- and Re-Rollers
- Respective motors
- And a control board

Because an old inkjet printer was at hand and the wheel did not need to be reinvented for a proof of concept it was taken apart to reuse its paper feeding mechanism. The original system uses brushless dc-motors with optical encoders for positional resolution. Since the goal was to control all parts of the machine with the same software interface, which interacts with Marlin firmware, those mechanisms were discarded. In exchange a Vextra PH265-01 stepper motor was mounted. At last the design was modified to be able to sit on top of the build plate, while being secured against sliding. The complete design can be seen in Figure (14).

Due to the gear ratio the steps per mm of the stepper driver needed to be redefined. Inside marlin, the steps per unit input are defined by the step pulses send to the stepper motor. Therefore:

- Stepper motor step angle: $1.8^\circ \Rightarrow 200 \text{ steps/rev}$
- Micro stepping: $1/16$
- Small gear: 20 teeth

- Big gear: 50 teeth
- Gear ratio: $\frac{20}{50}$
- Rod diameter: 8 mm $\Rightarrow$ circumference: 8π mm

Steps per mm:

$$\text{Steps per mm} = \left(\frac{200 \cdot 16 \cdot \frac{20}{50}}{8\pi} \right) \approx 51$$

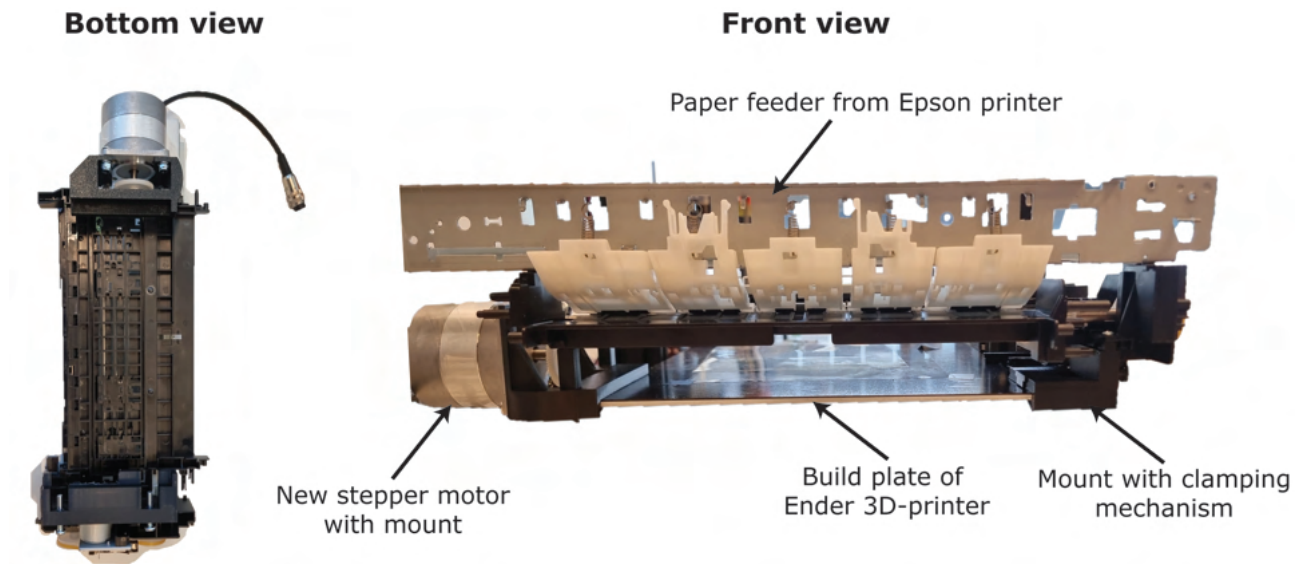


Figure 14: Assembly of paper feeder

Similarly the De-and Re-Rollers were designed by simplifying the approach of the Calisto. A single roller is then formed by the assembly of 2 frame pieces that mount the stepper motor on one side and a bearing on the other side. Connected by respective adapters and a PVC tube they form a free rolling system. For additional support, the assembly uses two rods at bottom (Figure (15)), that counteract the deformation due the torque on the system created by the motor. The technical information for the rollers can be found in the Appendix (44).

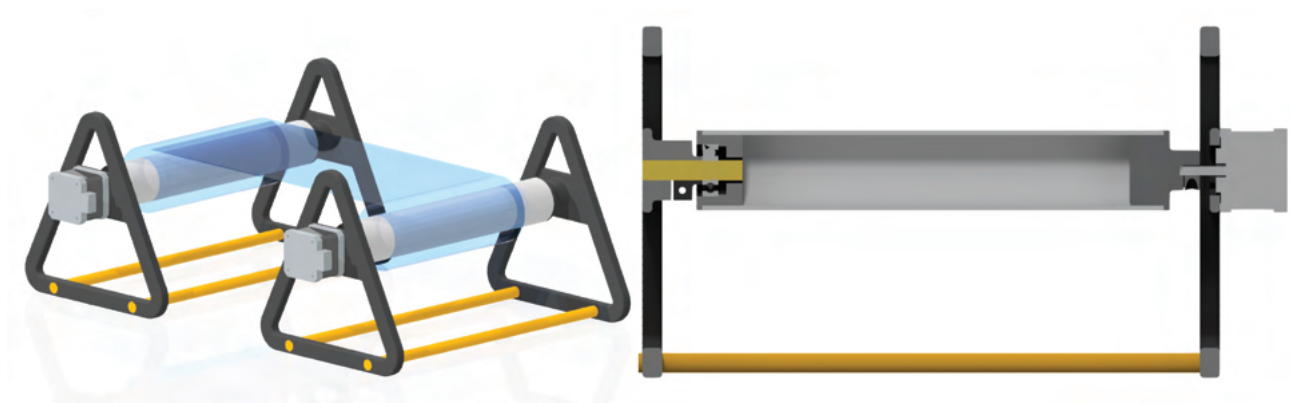


Figure 15: Assembly of rollers and a respective section view

The De-Roller side of the mechanism could then be loaded by feeding it from another roll. As this project

works with creating optodes the substrate is required to be translucent and non-fluorescent. Fortunately the Solar Lab at SDU is equipped with rolls of PET, which is a material that fits the requirements and was validated by use in the previous project.

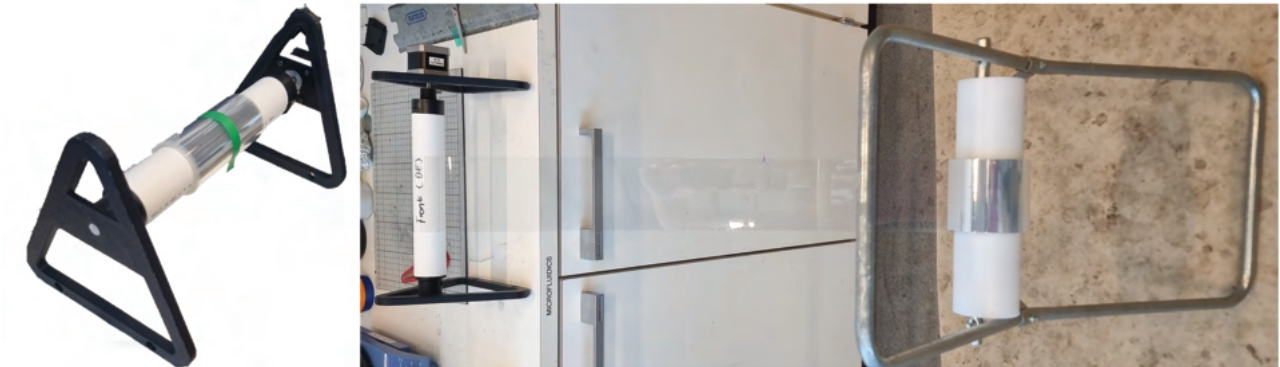


Figure 16: Loading the De-Roller before the spotting process

5.3 Syringe Pump

Working with low viscosity liquids requires special handling in the fluid path. Compared to high viscosity liquids, like glue or silicone, the capillary forces are not greater than the hydrostatic pressure, or would require a very small nozzle diameter to overcome them. As per previous work analysis solution of polystyrene in toluene in a concentration of 10% w/v is used, which has a viscosity of 5-10 cP while the nozzle is between 0.2 and 0.4 mm the solution would leak as the capillary forces are not greater than the hydrostatic pressure, which were shown during testing.

To overcome this problem the reservoir has to be sealed off. The liquid leaking then creates a vacuum pressure in the reservoir, that counteracts the hydrostatic pressure, keeping the liquid from leaking out of the nozzle. This is referred to the equilibrium point. On the other hand, this means after shooting a droplet, there will be negative pressure gradient in the reservoir, which will cause the liquid to be sucked back into the nozzle. This is referred to the backflow.

Therefore, the reservoir has to be modular to always keep the equilibrium point in the reservoir. This can be done by using a syringe as the reservoir, which then can then be precisely moved by a stepper motor, creating a syringe pump. Using laboratory syringe pumps, like the [NE-1000](#) would work but add another device to the setup which needs to be interfaced, and implemented into the software. Therefore, a custom syringe pump was designed, which can be mounted on the printer and utilizes a NEMA 17 42-34 stepper motor. As SHi et. al [27] showed, the pressure in the reservoir and the backflow can be used as another metric for controlling the droplet size.

A syringe pump was then designed and fitted with an adapter to mount to the Ender's frame as seen in Figure (17). The dimension are found in the Appendix (46).

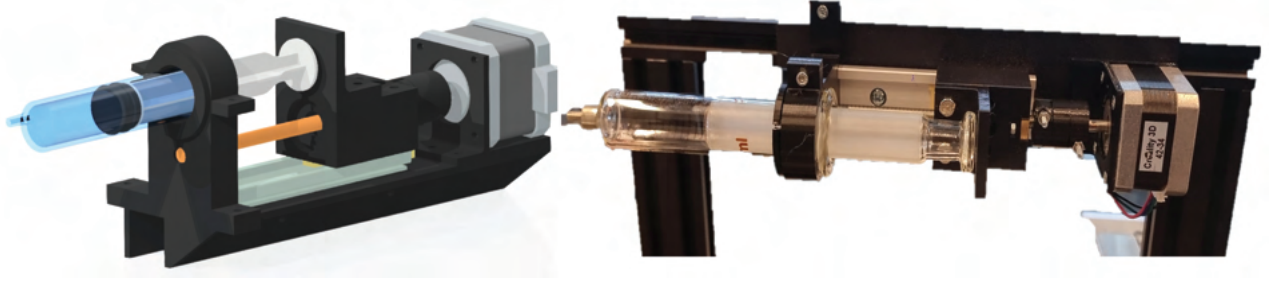


Figure 17: CAD and mounting of syringe pump

A syringe pump converts the rotary motion of a stepper motor into linear displacement of a syringe plunger via a leadscrew. In this setup:

- **Stepper motor:**

- Step angle: 1.8°
- Microstepping: $1/16$
- Steps per revolution:

$$N_{\text{rev}} = \frac{360^\circ}{1.8^\circ} \circ 16 = 3200 \text{ steps}$$

- **Leadscrew:**

- Diameter: 6 mm
- Pitch: 1 mm/rev (each full turn advances the plunger 1 mm)

- **Syringe:**

- Inner diameter: 17.2 mm
- Cross-sectional area:

$$A = \pi \left(\frac{D}{2} \right)^2 = \pi \left(\frac{17.2 \text{ mm}}{2} \right)^2 = \pi \cdot (8.6 \text{ mm})^2 = 73.96\pi \text{ mm}^2 \approx 232.3 \text{ mm}^2$$

A linear advance of the plunger by 1 mm displaces:

$$V_{1 \text{ mm}} = A \times 1 \text{ mm} = 232.3 \text{ mm}^3 = 232.3 \mu\text{L} \quad (1 \text{ mm}^3 = 1 \mu\text{L}). \quad (1)$$

Thus, each full revolution of the leadscrew (1 mm advance) delivers

$$V_{\text{rev}} = 232.3 \mu\text{L}.$$

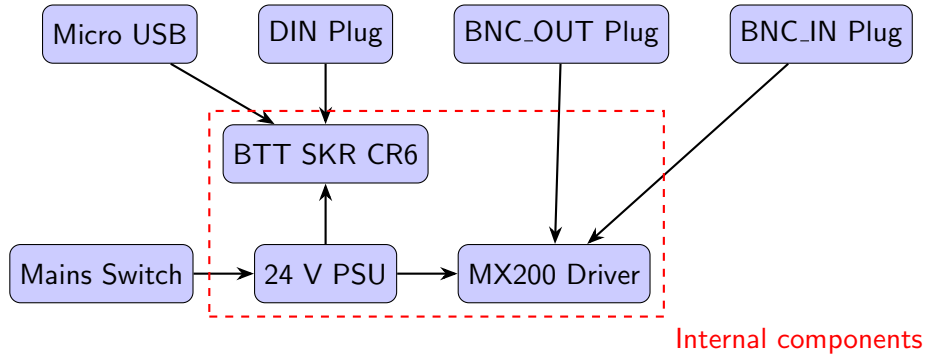
Since there are $N_{\text{rev}} = 3200$ steps per revolution:

$$V_{\text{step}} = \frac{V_{\text{rev}}}{N_{\text{rev}}} = \frac{232.3}{3200} \mu\text{L} = 0.0725 \mu\text{L} \approx 73 \text{ nL}. \quad (2)$$

As shown in (6) an average with droplet with 800 nm footprint has a volume of: 80 nL. To enable continuous spotting, the syringe pump must supply 1 step per droplet and 2 steps every 12th droplet.

5.4 Control box

In order to avoid having electronic components sitting openly, creating risk of shorting, damaging and a general cable chaos a control box was designed that acts as a housing for the Mx200 piezo driver from PiezoDrive, the R2R control board (Bigtreetech SKR CR6-SE) and their shared 24v 10A power supply. The added mains switch also allows to turn the system on and off in a safe manner.



The control box (Figure (18)) was then designed to stand next to the printer as a standalone unit. Starting by sketching all components (high resolution Appendix (41)) that would have to interact with each other and following with an assembly of the electronic components, giving them appropriate spacing. The box could then simply be build around them.

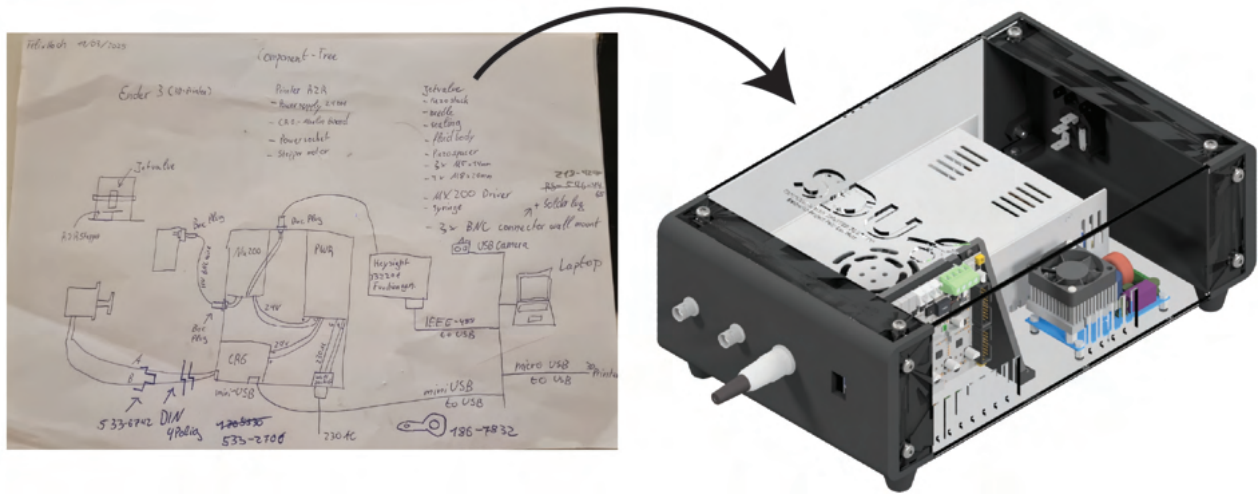
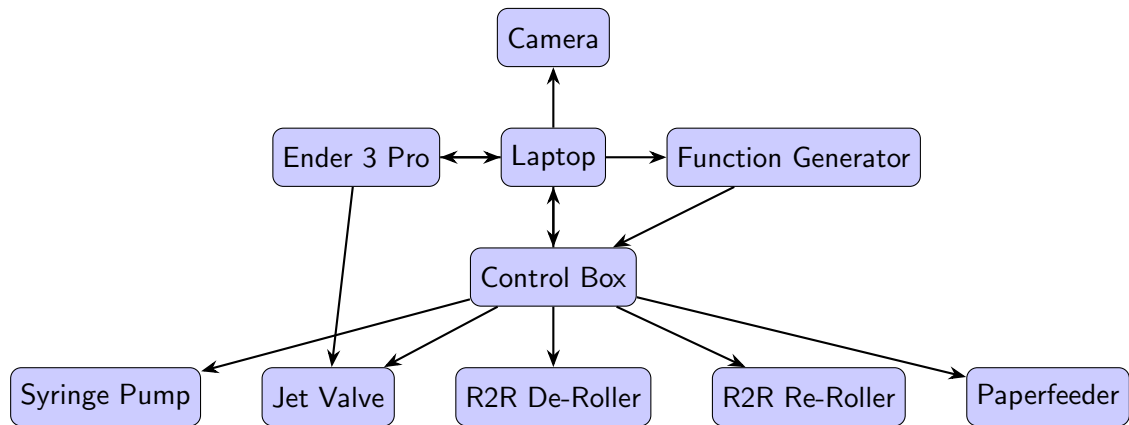


Figure 18: Control box development from sketch to assembly

5.5 Component fusion

Finally, after having introduced all components and subsystems, the complete connectivity of the system can be shown as:



6 Programming

The programming of the software was separated into sub-points. At first, scripts had been developed to control each individual device and once they had been tested, all were combined using a Graphical User Interface (GUI). Inside the GUI all settings of the spotting process could be manipulated and the machine controlled without additional touching of python scripts.

6.1 Jet-Valve interface

In order to control the Jet-Valve several components are needed to work in a chain. Using the top down approach as:

1. The needle requires 200-300 μm movement for accelerating droplets
2. The lever amplifier mechanism amplifies 25 times, requiring a 10 μm displacement actuator
3. The piezoelectric stack requires a driving voltage of 0-75VDC for a displacement range of 0-11 μm
4. The piezo-amplifier outputs 0-200VDC, requires 0-10VDC input and amplifies 20 times
5. The function generator must be limited to output a max peak voltage of 3.75VDC corresponding to the max 75VDC to not overload the piezo
6. The function generator must be triggered by GPIB signals for true Drop-On-Demand(DOD) behavior

As a measure of safety and because direct voltage control is not required, as control systems like PID for droplet sizing have been out of scope due to time constraints, all voltage controlling settings have to be set manually by the operator. This ensures that no output, that could potentially harm the piezoelectric stack is being generated.

Hence the task of the software is to generate signals that can turn the output of the function generator "ON" and to send a trigger signal, which activates a predefined burst signal.

Communication between the software and the function generator is handled via the General Purpose Interface Bus (GPIB). This method was chosen as it is a legacy standard for laboratory environments and comes with high reliability, robustness and shielded cables. As the IEEE-488 (GPIB) standard cables are expensive (1000€++) the communication protocol could have been switched to USB or Ethernet, if no cable would have been found in house.

Implementation for both GPIB and USB works the same on the software side as Python provides the communication library PyVisa with straightforward approaches to connecting and sending the Standard Commands for Programmable Instruments (SCPI):

```

1 import pyvisa
2
3 rm = pyvisa.ResourceManager()
4 if COM == "GPIB":
5     address = "GPIB::10" # Function generator GPIB address
6 else:
7     address = "USB0::0x1234::0x5678::INSTR" # Example USB address
8 jet_valve = rm.open_resource(self.address)
9 jet_valve.write("OUTP ON")
10 jet_valve.write("TRIG") # sending a single burst signal resulting in a DOD droplet
11 jet_valve.write("OUTP OFF")
12 jet_valve.close()

```

Listing 1: Pseudo code: Communication with Function Generator

Inside the GUI the functionality can then be directly accessed.

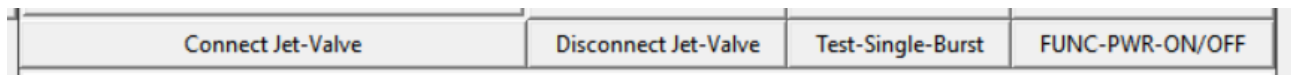


Figure 19: GUI field for Jet-Valve control

6.2 Printer interface

Similarly to the Jet-Valve, it is possible to directly send G-Code commands to a printer using Python's pySerial library. Here the commands are converted to binary which can be translated by the printers firmware.

```

1 import serial
2
3 connection = serial.Serial(
4     port = "COM3", # Example COM port, where the printer is detected
5     baudrate = 115200,
6     timeout = 1
7 )
8 command = "G28 X0 Y0 Z0" # Home all axes
9 connection.write(command.encode())
10 connection.flush()
11
12 response = connection.readlines(1) # Handling the printers response
13
14 connection.close()

```

Listing 2: Pseudo code: Communication with 3D-Printer

Marlin configuration

In order to make a standard 3D printer like the used Ender 3 Pro capable of communicating with other devices for real DOD behavior, the Marlin firmware had to be customized. By enabling the "get current position" M114_REALTIME support inside the configuration_adv.h of Marlin, probing the current position of the stepper motor steps would become possible. Without this change, M114 would be queued into the G-Code planner, hence only replying once a move had been finished. Similarly, the printer "safe homing" has to be disabled to allow Z-probing with an un-homed y-axis while also disabling the software endstops, so infinite y axis movement would become possible. After those changes and adjustment of the stepper

motor current to fit the new stepper motor used on the R2R add-on the two systems could be combined and calibrated.

It was initially planned to run the paper feeder of the control box. Unfortunately the BTT SKR CR6 used in the control box ran out of long time support in 2021 and the new Marlin features required came with Marlin 2.1+ updates in 2022, which was only found after implementation. This means, that it is not possible to get realtime feedback of the rollers and resulted in the De- and Re-Rollers being implemented in relative positioning mode. To work around this issue the paper feeder was then connected to the Ender 3 Pro's y-axis, which was compatible with realtime support.

6.3 Computer vision determining footprint size

In order to use a camera for the real-time analysis of the droplet footprint size several steps of image processing had to be done.

6.3.1 Deskew calibration

At first the skewing of the image had to be corrected. Due to space constraints the camera was mounted at an 40° angle relative to the jetting direction. This would result a horizontal skewed perspective of the substrate, introducing an elliptical deformation of the droplet footprint. To create accurate results, the perspective must be deskewed.

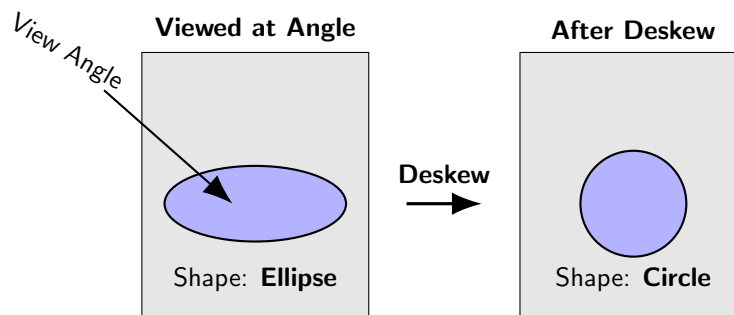


Figure 20: Visualization of elliptical error and Deskew transformation

There are methods to find circles at any point of the image and directly calculate the required homography transformation like homography estimation [7] and apply the deskewing directly to each individual circle. But due to the consistent position of the droplets and camera, as images are taken with the spotted droplet being in the center of the frame, resulting in the same skewed perspective at every analyzed footprint a simpler approach was chosen.

Therefore, the deskew-calibration-paper is loaded into the machine and the camera height is adjusted so that the resulting spotted droplet lands the vertical and horizontal center of the image. The deskew-calibration-paper's calibration window is then aligned with the center of the camera so that the droplet will land on the center of the cross and an image is taken. A calibration script is then run, which finds the four corners of the square and calculates the homography transformation matrix. This matrix is then used to calculate the distortion matrix, which is used to deskew the image of the calibration window. At last, the resulting deskewed image is analyzed in a vector graphics program to validate the back-transformation to a perfect square Figure 21 9).

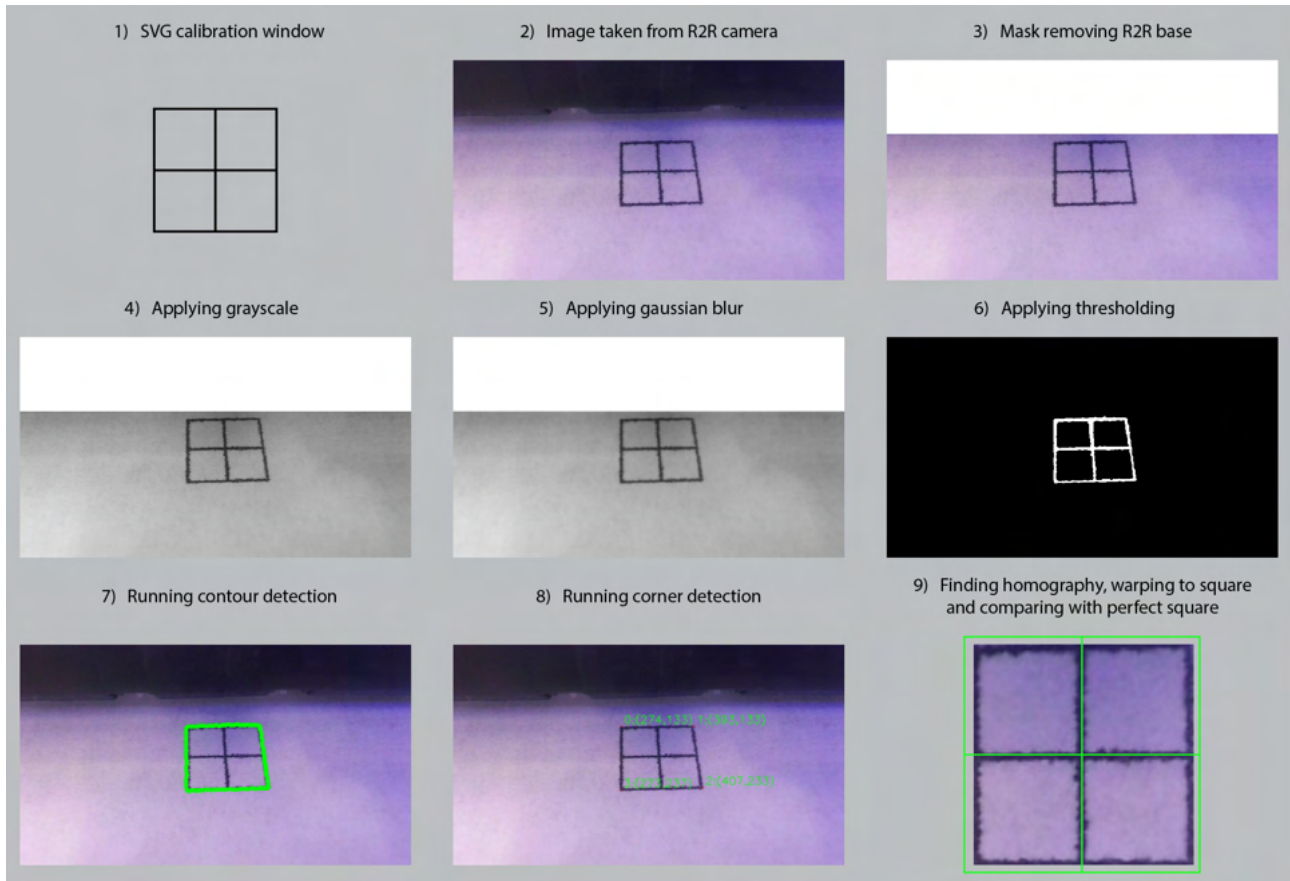


Figure 21: Deskew calibration process

This is implemented in the code by:

```

1 import cv2
2 import numpy as np
3 import json
4
5 img = cv2.imread("deskew_calibration.jpg")
6 # corner points of the ignored area mask
7 m_c_TL = (0, 0)
8 m_c_BR = (1920, 400)
9 cv2.rectangle(img, m_c_TL, m_c_BR, (255,255,255), -1)
10 # convert the image with the mask to gray scale
11 gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
12 blurred = cv2.GaussianBlur(gray, (5, 5), 0)
13 _, bw = cv2.threshold(blurred, 100, 255, cv2.THRESH_BINARY_INV)
14 # Find the largest 4 point contour
15 contours, _ = cv2.findContours(bw, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
16 src_pts = findCornerPoints(contours)
17 # size to which the deskewed image will be resized
18 side = 300
19 dst_pts = np.array([[0,0], [side,0], [side,side], [0,side]], dtype=np.float32)
20 M = cv2.getPerspectiveTransform(src_pts, dst_pts)
21 deskewed = cv2.warpPerspective(img, M, (side, side))
22 cv2.imshow("Deskewed", deskewed)
23 # saving the corner coordinates into a json file for later use
24 labels = ['TL', 'TR', 'BR', 'BL']
25 corners_dict = { lbl: [int(pt[0]), int(pt[1])] for lbl, pt in zip(labels, src_pts) }

```

```

26
27 with open('detected_corners.json', 'w') as f:
28     json.dump(corners_dict, f, indent=2)
29
30 cv2.waitKey(0)
31 cv2.destroyAllWindows()

```

Listing 3: Pseudo code: Deskewing the calibration window

The underlying homography transformation operation can be described as:

We want to find a 3×3 matrix

$$M = \begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & 1 \end{pmatrix}$$

that carries each source corner (x_i, y_i) to a destination corner (u_i, v_i) in homogeneous coordinates:

$$\tilde{p}'_i = \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} \sim M \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} \quad (i = 1, \dots, 4),$$

where “ $\sim$ ” means equality up to an arbitrary scale. Writing

$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = \lambda_i \begin{bmatrix} m_{11}x_i + m_{12}y_i + m_{13} \\ m_{21}x_i + m_{22}y_i + m_{23} \\ m_{31}x_i + m_{32}y_i + 1 \end{bmatrix},$$

and eliminating the multiplier λ_i gives the two scalar equations

$$\begin{cases} u_i (m_{31}x_i + m_{32}y_i + 1) = m_{11}x_i + m_{12}y_i + m_{13}, \\ v_i (m_{31}x_i + m_{32}y_i + 1) = m_{21}x_i + m_{22}y_i + m_{23}. \end{cases}$$

This yields $2 \times 4 = 8$ linear equations in the eight unknowns

$$\{m_{11}, m_{12}, m_{13}, m_{21}, m_{22}, m_{23}, m_{31}, m_{32}\}.$$

Solving this system (e.g. via Gaussian elimination) produces $M = \text{getPerspectiveTransform}(\text{src_pts}, \text{dst_pts})$ in OpenCV.

Once M is known, each output pixel (u, v) in the $W \times H$ rectified image is sampled by the inverse map:

$$\text{warped}(u, v) = I(M^{-1}[u, v, 1]^\top), \quad 0 \leq u < W, \quad 0 \leq v < H,$$

where $I(x, y)$ denotes the source image's intensity at (x, y) (with bilinear interpolation). OpenCV's `warpPerspective(frame, M, (W, H))` implements exactly this inverse-mapping plus interpolation.

Once successful deskewing was achieved, the corner coordinates are saved and can be used inside the main program to apply a deskewing filter, for a image, on which later image processing will be done.

Further image processing

After reshaping the image, we can apply filters to see only the droplet at the center of frame of the live video feed, e.g. where we expect a spot to appear. For this, we convert the image into gray scale, add a blur effect to remove impurities and apply a threshold mask, which will remove everything but the spots.

6.3.2 Real-time image spot analysis

Once the image is deskewed and the footprint is isolated, we can analyze the image to find the droplet footprint size and save it for background analysis. This allows the software to generate a report of the spotted grid and directly show the user the performance of each grid.

This is done by again looking for the contours of the processed image and assuming that the largest contour is the outside edge of the droplet footprint. That contour is then used to calculate the area of the droplet footprint from which the diameter in pixels can be calculated by:

$$d_{px} = \sqrt{\frac{4 * Area}{\pi}} \quad (3)$$

Which divided by the pixel per mm value gives us the diameter in mm.

$$d_{mm} = \frac{d_{px}}{pixel/mm} \quad (4)$$

At this point it is important to adjust the threshold value so the recognized contour fits the actual droplet footprint. For this purpose the circle detection returns the image with measured diameter and the circles outline.

The processed camera feed is therefore set up as:

```
1 import cv2
2 import tkinter as tk
3 from PIL import Image, ImageTk
4 import numpy as np
5
6 def update(self):
7     if self.camera_feed and self.camera_feed.vid is not None:
8         ret, frame = self.camera_feed.vid.read()
9         if ret:
10             # source points for deskewing found with the calibration window
11             src_pts = np.array([[855,473], # Top left
12                                [1124,471], # Top right
13                                [1153,702], # Bottom right
14                                [854,708] # Bottom left
15                                ],
16                                dtype=np.float32)
17             # destination points for the deskewed image
18             dst_pts = np.array([[0,0], [320,0], [320,320], [0,320]], dtype=np.
19                                float32)
19             M = cv2.getPerspectiveTransform(src_pts, dst_pts)
20             warped = cv2.warpPerspective(frame, M, (320, 320))
21             # image processing
22             gray = cv2.cvtColor(warped, cv2.COLOR_BGR2GRAY)
23             _, binary = cv2.threshold(gray, self.threshold_var.get(), 255, cv2.
24                                     THRESH_BINARY)
25             self.processed_img = binary
26             d_mm, processed_image = self.measure_droplet(self.processed_img, self
27                                                         .pixel_per_mm)
28             self.current_diameter = d_mm
29             self.canvas.add(processed_image)
30             self.parent.after(30, self.update)
31
32 def measure_droplet(self, img, pixel_per_mm):
33     contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE
34                                   )
35     cnt = max(contours, key=cv2.contourArea) # selects circle
```

```

33     area = cv2.contourArea(cnt)
34     vis = cv2.cvtColor(img, cv2.COLOR_GRAY2BGR)
35     cv2.circle(fit_circle on spot)
36
37     diameter = np.sqrt(4 * area / np.pi)
38     d_mm = diameter / pixel_per_mm
39     d_mm = round(d_mm,2)
40     text = f"Diameter: {d_mm}mm"
41     cv2.putText(Diameter)
42     return d_mm, img

```

Listing 4: Pseudo code: Real-time image spot analysis

The resulting image is then shown in the GUI and the diameter of the droplet footprint is displayed. For the spotting process the user can then choose to measure every droplet or only the first one of each row:

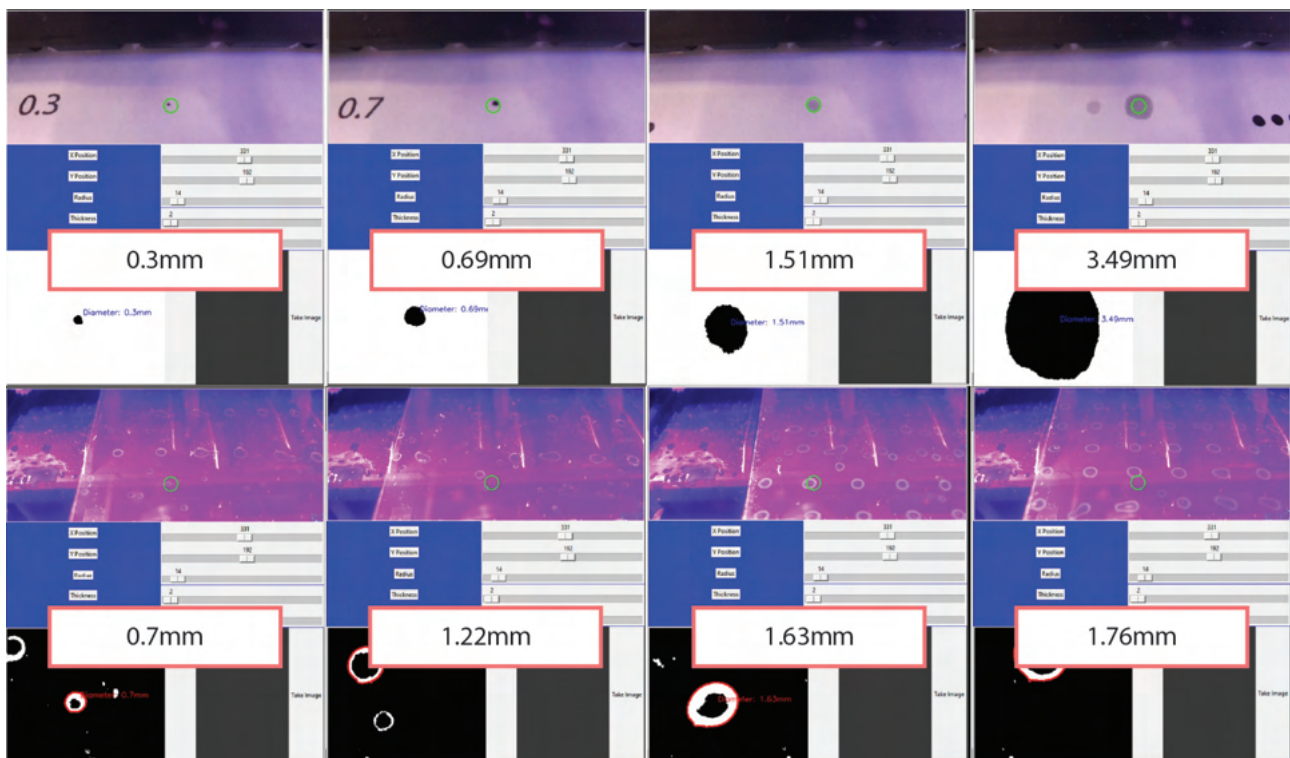


Figure 22: Realtime footprint measurement of calibration paper and spotted spots

6.4 Data curation

The live data creation process can be illustrated by Figure (23):

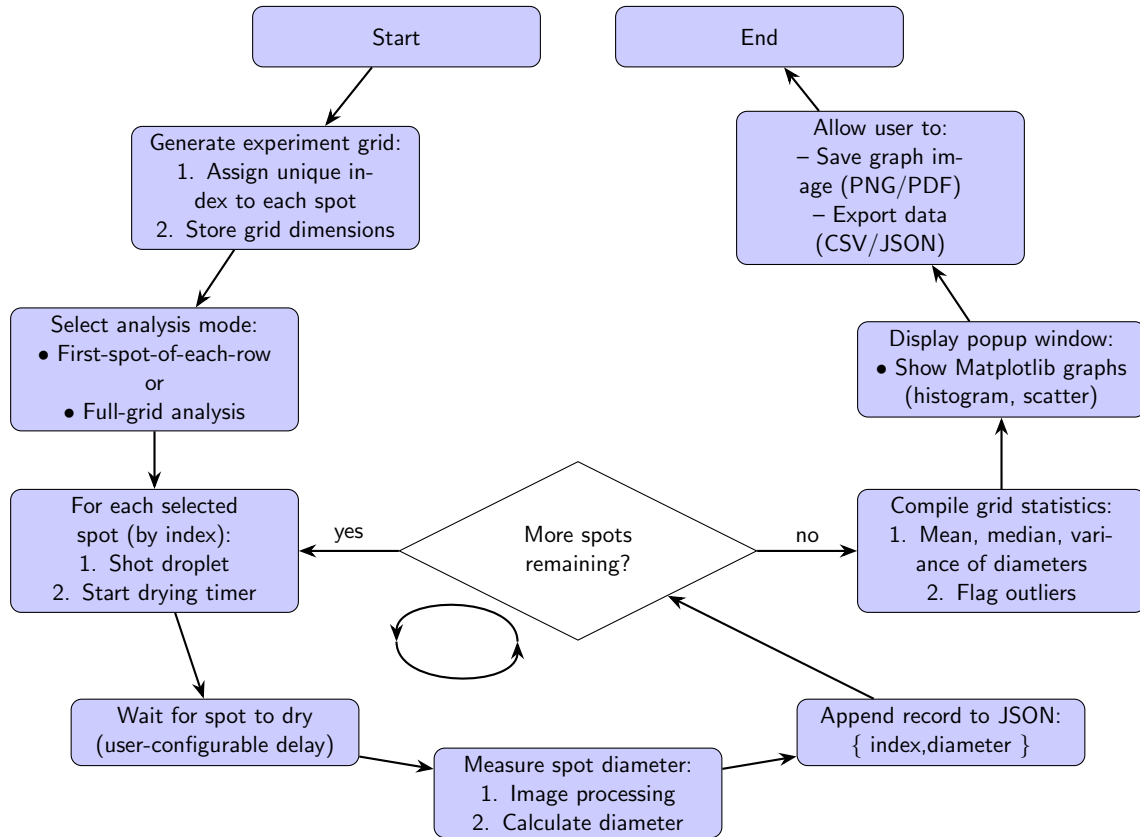


Figure 23: Flow diagram of footprint data curation process

Once the user selects the analysis mode, the software saves either no spot data, the first of each row or the data of all spots. As each measurement introduces a drying time to the process, the resulting long spotting times for large grids are to be considered. Hence, first of row measurement would be the recommended mode for performance tracking during normal operation.

The created output in full grid mode is, for example (Figure (24)):

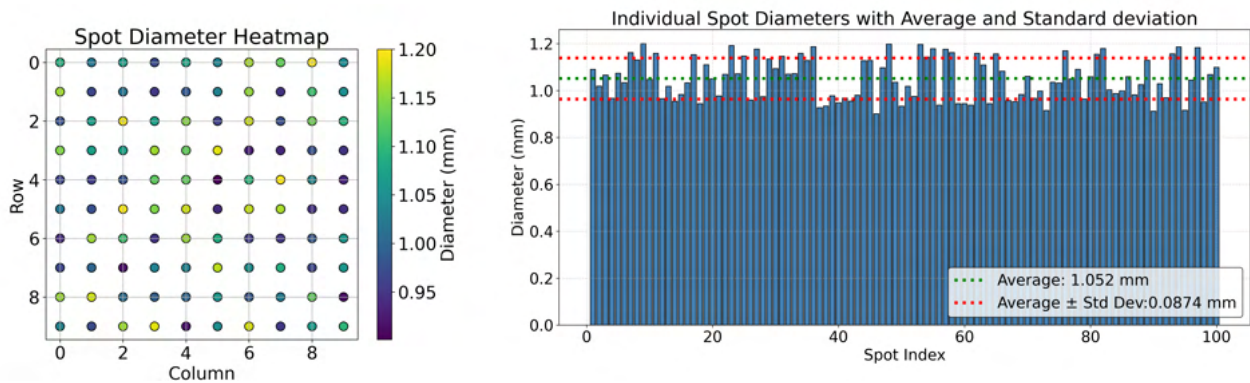


Figure 24: Sample output with generated diameter data

6.5 Desktop application

The software was developed using the Python programming language and the Tkinter library for the GUI. The GUI is designed to be user-friendly and allows the user to easily control the spotting process, adjust settings, and visualize the results. The main features of the software include:

- Ability to define grid
- Ability to see the spotting process
- Ability of spot by camera positioning
- Saving presets

The software is separated into two modes, namely the Build-plate-mode and the R2R-mode which come with their respective settings and configurations. The focus of the project was on developing the R2R-mode, but the Build-plate-mode allows the user to use the y-axis of the original printer, and hence create grid patterns at size of the complete build plate. The work flow for both is the same, except for manually choosing the dedicated R2R machine in R2R mode.

The user flow can be described by Figure (25):

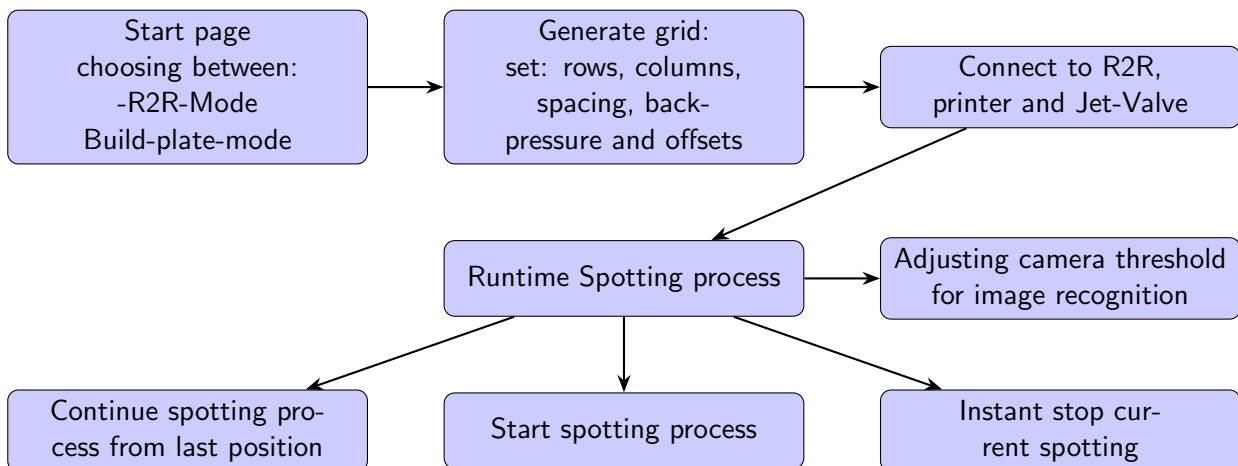


Figure 25: Desktop application user flow diagram

After the spotting process, the data curation workflow as previously described is then running to store the data of the footprints. An example state of the application looks like Figure (26):

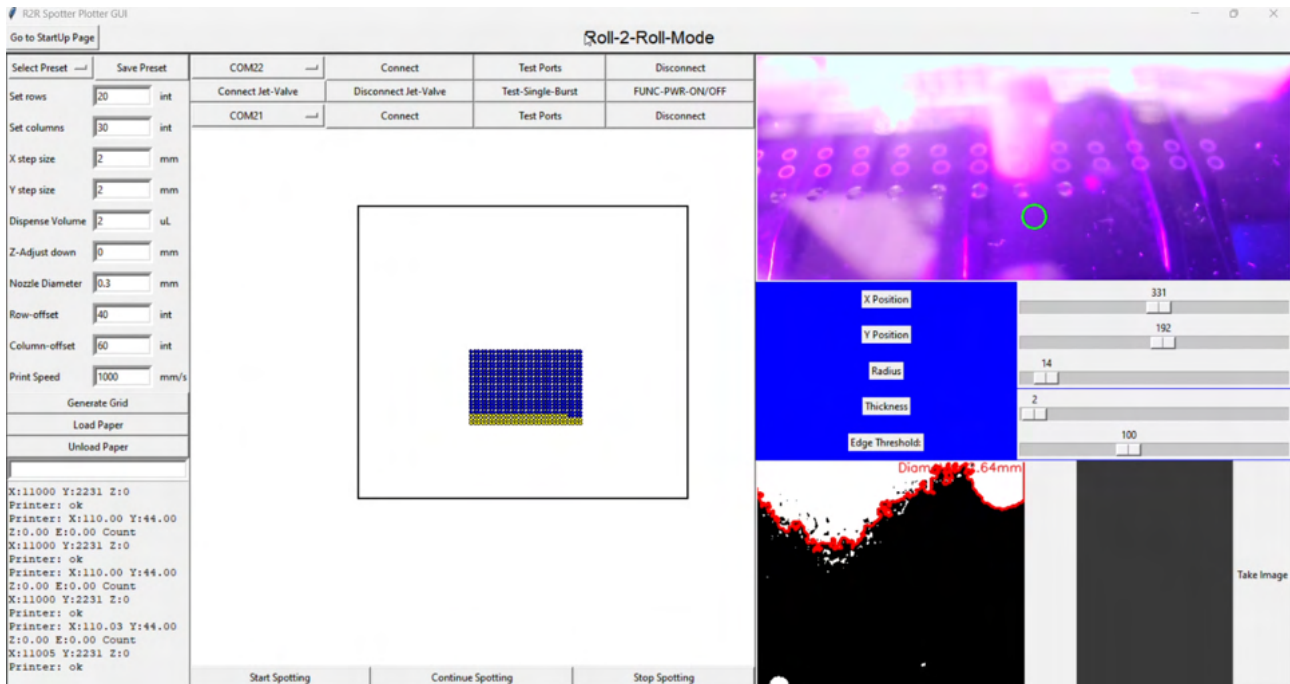


Figure 26: Example state after spotting process

The window in Figure (26) show cases a typical spotting process. On the left is the input frame, which defines the grid. Upon setting the parameters and clicking "Generate Grid" the grid is generated and placed inside the black square, which shows the relativity to the build plate of the Ender. The grid starts completely blue and after starting the spotting process each completed spotting action turns it's respective spot on the canvas yellow, giving the user live feedback of the progress.

One of the problems of live spot size detection becomes very clear, when looking at the camera in the top right of the window. The current active spot as well as the last 8 spots have not yet dried and not left a detectable footprint. This as well as the light source used for illumination must be put to consideration during the calibration phase.

Programming logic

The design pattern chosen for the development of the Graphical User Interface (GUI) and fusion with the backend was the Model View Controller (MVC). This pattern separates the front end from the backend and creates interfaces that handle the communication. The front end is therefore never directly accessing backend functions which makes the codebase more scalable and maintainable.

The codebase consists of three view scripts (Start, R2R and Build-plate), their view manager, the backend scripts for Jet-Valve and G-Code Serial communication handled by a device manager, the grid backend, the model manager, handling grid and devices and at last the main controller fusing all together. In addition, there are scripts for startup (main), drawing utils, the camera feed, the processed camera feed and a data curation helper. Visualizing this in a dependency flowchart (Figure (27)) gives:

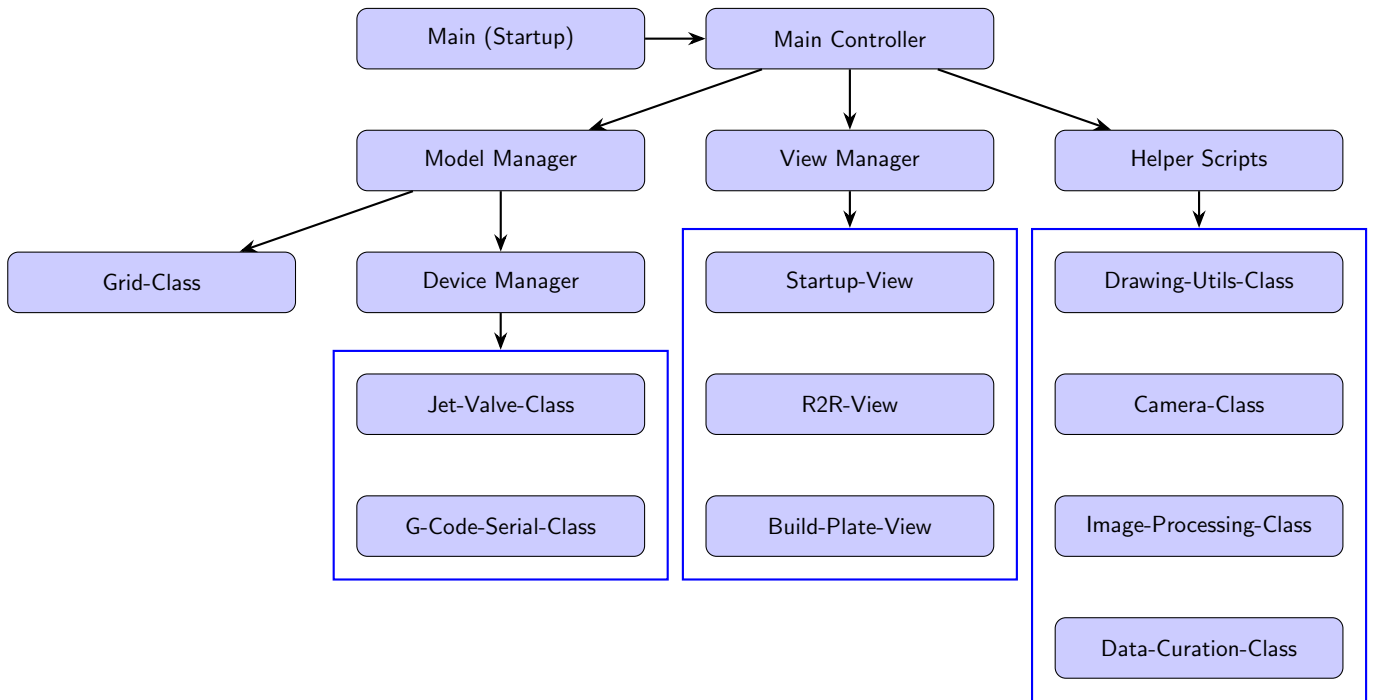


Figure 27: Desktop application interaction diagram

Inside the main controller the spotting logic is handled. At first, after the user has input the grid clicked generate, two functions are called. For one, the `send_grid_gcode()` which generates a coordinate for each spot. Those are then the x and y GO, which will be send to the printer. Everyone of those coordinates is then multiplied by the steps per unit, which was previously calibrated, and saved to an array of x and y REQUIRED, as those are the values, in real steps, which will be returned from the machine after requesting the realtime position.

After having generated those two 2D arrays, the spotting process is started. Dependent on R2R or Build plate mode a individual spotting process is choosen. The R2R spotting process can be illustrated as:

```

1 def r2r_spotting_process(self, grid, stop_flag):
2
3     printer = self.model_manager.device_model.r2r_serial
4     r2r = self.model_manager.device_model.r2r_addon_serial
5     save_mode = self.save_mode
6
7     for row in range(self.current_row, grid.rows):
8         for col in range(self.current_col if row == self.current_row else 0, grid.
9             columns):
10             # stop flag raise will pause the spotting process but saves the current
11                 position
12             if stop_flag.is_set():
13                 self.current_row = row
14                 self.current_col = col
15                 return
16
17     number = 1 # Number of commands to send at once
18     self.r2r_send_next_gcode(number, grid, printer, r2r)
19     # Ensure snake spotting drawing
20     col = self.snake(row)
21
22     # set required coordinates for the current grid index

```

```

21         try:
22             x_req, y_req = self.grid_req[self.grid_index]['x_req'], self.
                grid_req[self.grid_index]['y_req']
23         except IndexError:
24             print("No more coordinates to process")
25             break
26
27         # requesting realtime position response
28         while True:
29             if stop_flag.is_set():
30                 self.current_row = row
31                 self.current_col = col
32                 print(f"Stopping at position: row {row}, col {col}")
33                 return
34             printer.send_command("M114 R\n")
35             threading.Event().wait(0.001) # dont clog the memory of the printer
36
37             # checking for response
38             printer_line = self.device_model.printer_response_queue.get()
39             printer_xyz = self.return_xyz(printer_line)
40
41             tolerance = 5
42             if printer_xyz is not None:
43                 if x_req-tolerance <= printer_xyz[0] <= x_req+tolerance and y_req
                    -tolerance <= printer_xyz[1] <= y_req+tolerance:
44                     print("X is good\n")
45                     # Spotting process
46                     self.make_burst_spot()
47                     plot.save_spot_diameter_to_json(self.commands_send, row, col,
                        mode)
48                     # Update the canvas
49                     recolor_circles(
50                         self.canvas,
51                         self.model_manager.get_grid(),
52                         row, col
53                     )
54                     break
55             try:
56                 self.grid_index += 1
57             except IndexError:
58                 break
59         # Create Data plots
60         plot.generate_diameter_graph()
61         plot.generate_diameter_stats_graph()
62         # Reset position after completion
63         self.reset(row,col,grid_index,command_send)
64
65     def r2r_send_next_gcode(self, amount, grid, printer, r2r):
66         print_speed = grid.print_speed
67         for number in range(amount):
68             number += 1
69             try:
70                 command2 = f"G0 Y-{grid.dispense_volume} Z{self.commands[self.
                    commands_send]['Y_go']} F{print_speed}\nG92 Y0\n"
71                 r2r.send_command(command2)
72                 command = f"G1 X{self.commands[self.commands_send]['X_go']} Y{self.
                    commands[self.commands_send]['Y_go']} F{print_speed}\n"
73                 printer.send_command(command)
74                 self.commands_send += 1
75             except IndexError:
76                 print("No more commands to send")

```

Listing 5: Pseudo code: R2R spotting process

As seen in the code (Listing (6.5) Line 35) the application uses threading. In order to have smooth operations, fast loading and task independent interactivity there are 4 threads setup, namely the application thread, that handles the user interface and updating of buttons, inputs and the canvas. The camera thread that updates the camera and processed camera, the spotting process thread that runs the spotting process as long the stop flag is not set and at last the printer response listener thread, which is responsible for listening to the position responses of the M114 REALTIME position request and putting the newest one into the current position queue. Here it is important to notice, that the queue is not actually a queue, but rather a process of overwriting the last value, as a real que introduces handling delays into the system. Of similar importance is to limit the response listener to only read one line at a time, as the system would otherwise wait for more incoming lines, which introduces a 5 seconds delay, hence bad for realtime operations. Limiting the listener to 1 results in instant transmission to the que after receiving data.

The full code base which at this point consists of 2526 lines of code can be found under [R2R-MVC](#).

7 Experimental calibration

7.1 Checking compatibility of fluid path

Because the working fluids of the project are polymer solutions dissolved in toluene, the compatibility of the components in the fluid path had to be validated. For that the materials can be checked up and even if they are prone to dissolve, a simple test can show the timescale of that process, as single use components could be utilized.

The chem lab provided the project with glass syringe, luer-lock dispensing tips and [FEP-Tubing](#) which had been rated stable for use with toluene by the manufacturer . The fluid inlet on the other hand had to be custom made and three different materials had been tested. For one the PETG FDM filament and for the other two the SLA resins FORMLABS RIGID 10K and FORMLABS CLEAR, for which only [PETG](#) is stated stable in toluene and both other have not listed information about compatibility.

Therefore, both had been fully submerged inside pure toluene and observed for the following week, which resulted in no visible dissolving and within a scratch test, the hardness also had not changed to a noticeable amount. Hence, the project planned on continuing with the RIGID 10k, which failed, after the part snapped due to unwanted tension created by the clamping screw biting into the wall, because of wrong tolerances. To proceed with the experiments the CLEAR resin inlet was then used.

Creating a custom sealing

After initial testing with mineral oil, it was seen, that the DIY'ed teflon disc sealing was not tightly sealing and a liquid leaking phenomena like described by Wu et. al [9] were seen. With using mineral oil as the working liquid, this fact was not problematic and grid spotting could continue unhindered. After switching the working liquid to the toluene polymer cocktail, the actual impact of this design flaw became evident. The leaking cocktail would work itself up the needle assembly and dry after reaching air inside the valve body. As the needle would move back down, the dried polymer got pushed back into the needle assembly, forming a blockage and resulting in the getting stuck, breaking the mechanism.

As this resulted in mechanical failure of the system, a solution had to be found in order to proceed any further with the project. Although not compatible with toluene, a custom rubber sealing had then been

manufactured out of a rubber disc. This resulted in a perfect seal, although long term compatibility is not given (38).

7.2 Calibration of R2R-Rollers

As the rollers have been custom made, there would not be presets on how their movement in millimeters relates to the steps or micro steps done by the stepper motor. This creates the requirement for calibration, to reliably feed substrates at demanded steps in mm.

Two approaches can be used for this:

- Relating outside diameter to micro-steps analytically
- Experimentally adjusting steps/mm

The experimental calibration (Figure (28)) was chosen, because it provides a direct measurement of the output performance and would be needed to validate the calculations from the analytical approach anyway. Later on the analytical approach is then calculated to find its prospective error in comparison.

The calibration goes as follows.

1. Request a linear movement of length: $L_{\text{req}} = 100 \text{ mm}$
2. Draw a line during the movement
3. Measure the actual length of the line: L_{true}
4. Calculate the ratio:

$$R = \frac{L_{\text{req}}}{L_{\text{true}}}$$

5. Multiply the current steps per mm by the ratio:

$$\text{Steps}_{\text{new}} = \text{Steps}_{\text{current}} \times R$$

6. Repeat the process until $L_{\text{true}} \approx L_{\text{req}}$

Resulting in:

Iteration	L_{req} [mm]	L_{true} [mm]	Current Steps/mm	Ratio $R = \frac{L_{\text{req}}}{L_{\text{true}}}$	Updated Steps/mm
1	100	78.5	80	1.274	101.92
2	100	101.7	101.92	0.983	100.18
3	100	≈ 100	100.18	---	---

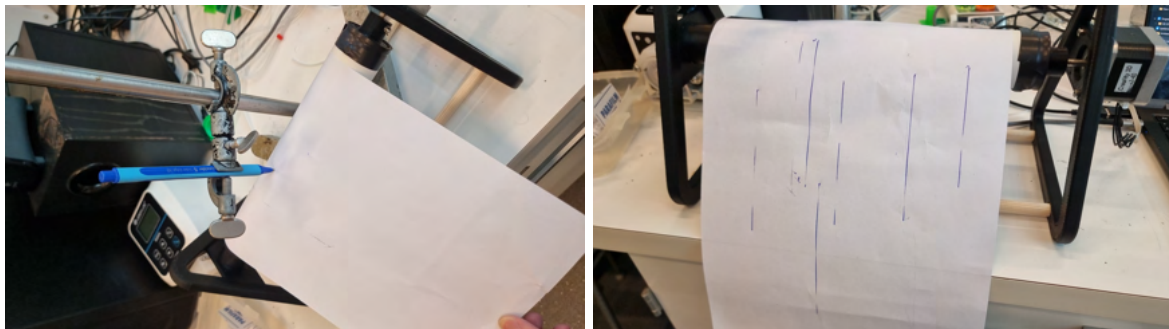


Figure 28: Calibration process for De- and Re-Rollers

The same process was repeated for the paper-feed resulting in 50.7 steps/mm for consistent 100mm movement.

Comparing this with the analytical method gives:

$$\text{Step angle} = 1.8^\circ \Rightarrow \text{Steps per revolution} = \frac{360^\circ}{1.8^\circ} = 200$$

$$D = 40 \text{ mm} \Rightarrow \text{Circumference of tube} = C = \pi D = \pi \times 40 \approx 125.66 \text{ mm}$$

$$\text{Steps per mm} = \frac{200}{C} = \frac{200}{125.66} \approx 1.592 \text{ steps/mm}$$

$$\text{Steps per 100 mm} = 1.592 \times 100 \approx 159.2$$

Steps per 100 mm ≈ 159

The difference in results can have two possible reasons. For one, the stepping angle could be different than advertised and second, the driver implementation could be influencing the results of the experimental calibration. Either way, as the experimental calibration, although in an iterative, but straight manner, achieves the required output, choosing it as the standard method is validated.

7.3 Analysis of droplet footprint

For all following analysis and experimental calibration of the setup it is important to precisely measure the footprint of the spotted droplets. In professional equipment like [SmartDrop](#) and [sciDrop Pico](#) high speed cameras observe the droplet during falling, making it simple to calculate volume. Because those setups only provide initial calibration before the spotting procedure and would add a new layer of complexity it was decided to use the resulting droplet footprint as the point of measure.

Having a camera running in parallel to the spotting head and analyzing, provides data of every spot before and after surface wetting and additionally after evaporation of any solvents. Later, this can then be used to find the true volume of the droplets after leaving the nozzle.

7.3.1 Calibration of the camera

The camera on the R2R Spotter Plotter has to be calibrated to use it for live detection of footprint sizes. To do that we can look at circles of known size, measure their diameter in pixels and then calculate the pixel per mm value, which can be used in live analysis of the spotted droplets.

To verify that the camera on the R2R Spotter Plotter is calibrated with the correct expected droplet sizes a calibration paper was created with an inkjet printer, printing circles drawn in vector format from 0.1mm to 3.0mm in 0.1mm steps. To rule out recursive errors induced by calibrating with wrong calibration circle sizes, the calibration paper was analyzed under an optical microscope with 10x magnification seen in Figure (29). Their size was then measured in pixels and compared to the pixel per mm value of an object with known size e.g the pixel per mm value of a caliper, which at this point is assumed to be close to 100% accurate.

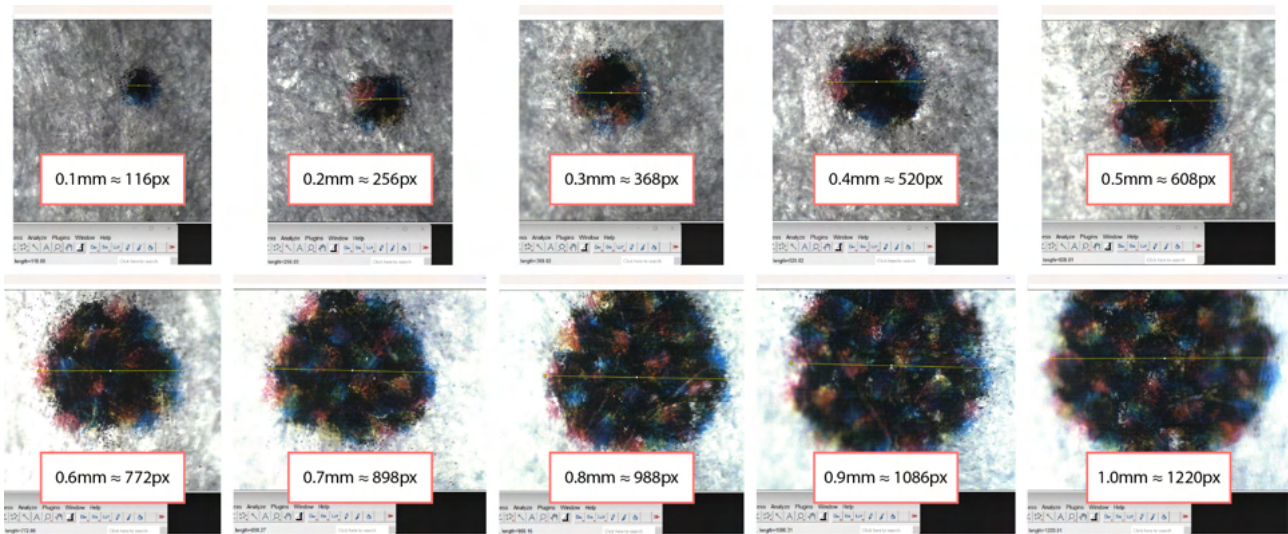


Figure 29: Measurement of ink-jet spot pixels under X10 microscope

Samples 1–5			Samples 6–10		
Diameter (mm)	Pixels	Rounded (mm)	Diameter (mm)	Pixels	Rounded (mm)
0.1	118	0.10	0.5	608	0.50
0.2	256	0.21	0.6	772	0.64
0.3	367	0.30	0.7	898	0.74
0.4	520	0.43	0.8	988	0.82
0.4	520	0.43	0.9	1086	0.90

Table 4: Pixel Measurements and Converted Sizes at 10x Microscope Magnification (1212.20 px/mm)

There are several problems with this approach. For one, it can be clearly seen under the microscope (Figure (30)), that the inkjet printing of calibration circles does not provide reliant circular base circles with a sharp edge. For circles with diameters below 1.0mm the shape is rather rectangular or triangular with a blurry edge that does allow for precise measurement of the intended size, because the measurement ruler does not have a sharp edge measure from.

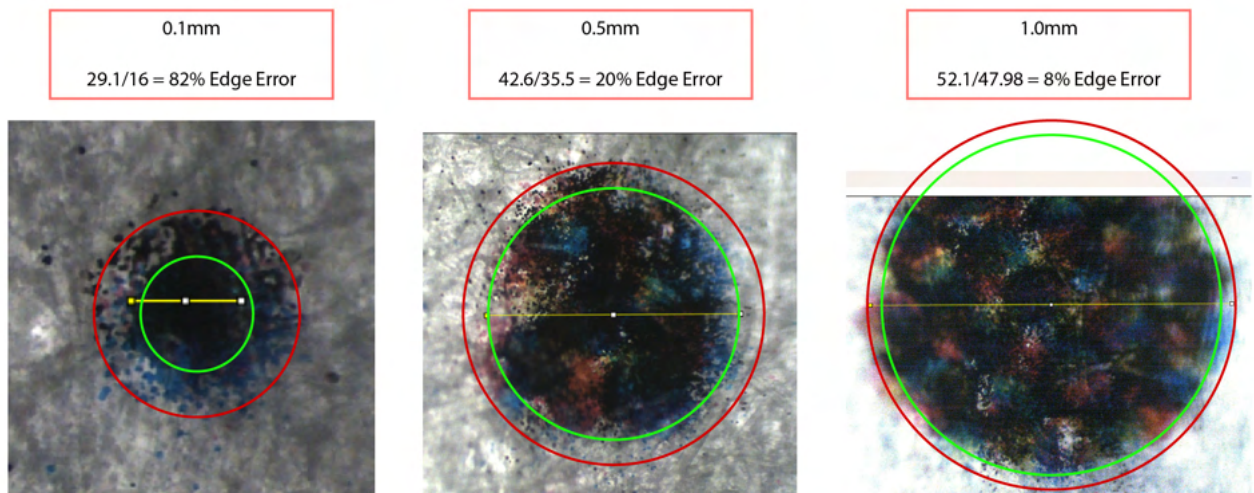


Figure 30: Error approximation of the calibration circles

Although not previously defined, it was chosen that the edge selection error band shall not exceed 10% during calibration, to reduce recursive error during live analysis. From this we can therefore conclude, that the ink-jet paper calibration method is not suitable for calibration of circles below 1.0mm.

Because the system is designed to measure the footprint size single droplets and those inherently should produce a circular, clean edge footprint while drying on the substrate, the next step of calibration is to calibrate the pixel per mm based on actual footprints spotted on R2R Spotter Plotter.

For that purpose a calibration pattern will be printed with the machine. During trial testing it was shown, the droplet size can be adjusted consistently by adjusting the stroke length. Hence, the burst spot method will be used, with respective different drive voltages. Therefore, the pattern will be created by, spotting multiple rows of 20 droplets and changing the voltage iteratively. The resulting footprint can later also be used for further analysis.

Testing was then done with a toluene solution, as this is the polymer cocktail used in the previous project, that after drying leaves a polymer footprint on the substrate. As the droplets had been getting bigger, this run was analyzed under X5 magnification 31.

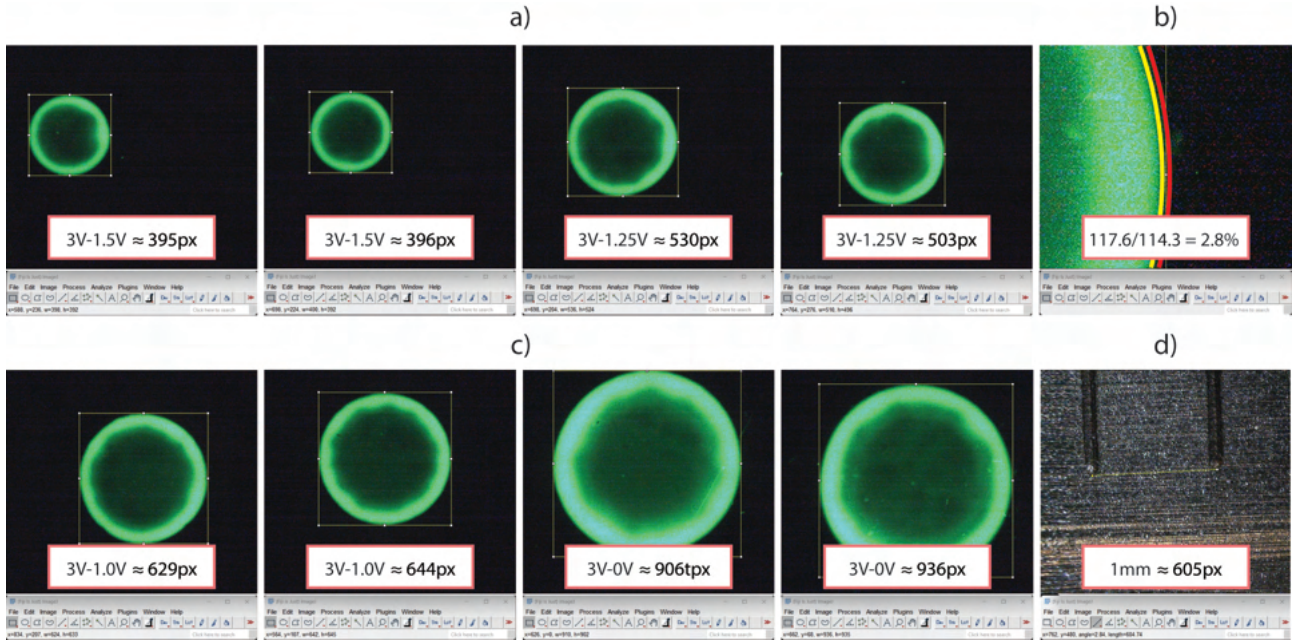


Figure 31: Measurement of spot pixels under X5 microscope, Figure b) showcases that the new calibration circles come with a satisfactory edge selection error. It can be seen how calibration of small footprint is better than with the ink-jet calibration paper because a clean edge can be grabbed. Figure a) and c) display the new measured spots, while Figure d) shows the calibration scale of the caliper.

Samples 1–4			Samples 5–8		
High Level - Low Level	Pixels	Rounded (mm)	Diameter (mm)	Pixels	Rounded (mm)
3V-1.5V	395	0.65	3V-1.0V	629	1.04
3V-1.5V	396	0.66	3V-1.0V	644	1.07
3V-1.25V	530	0.88	3V-0V	906	1.50
3V-1.25V	503	0.83	3V-0V	936	1.55

Table 5: Pixel Measurements and Converted Sizes at X5 Microscope Magnification (604 px/mm)

Knowing discrete footprint's with their respective diameter the camera calibration can continue through measuring exactly those. With an image captured by the camera mounted on the Jet-Valve and running a calibration script, that relates the captured circles diameters to the measured ones the pixel per mm value can be of that camera can be found. It is important to run the live measurement at the same height as the calibration images had been taken, as the pixel per mm are directly related to it.

7.3.2 Footprint against drive voltage

Now analyzing dependency of the droplet footprint to the driving voltages, 2 parameters, namely the high and low level voltage. Adjusting the high level results in direct change of the total stroke displacement of the needle, while adjusting the low level would introduce additional pretension of the system. That pretension induces a decrease of total stroke displacement but relative to the output nozzle. This can be described by Figure (32):

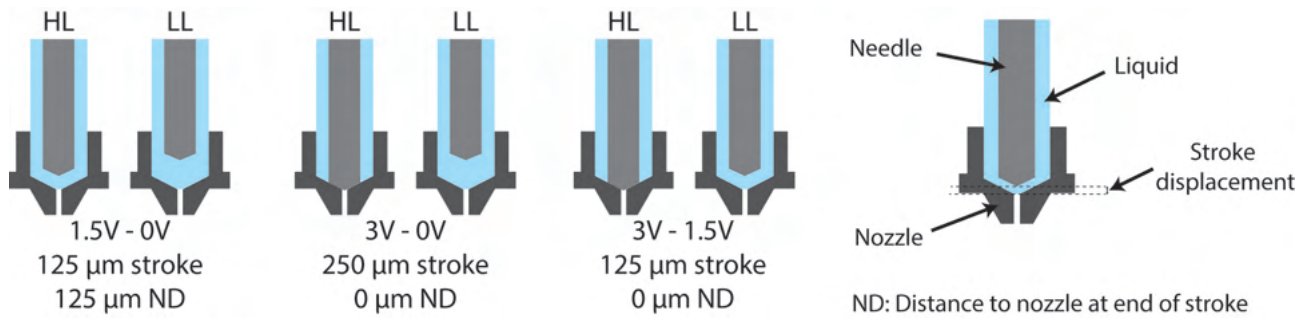


Figure 32: Drive voltage experiment description

As SHI et. al[27] already presented the direct relation between high level voltage and resulting droplet diameter, this experiment can focus on the impact of the low level voltage. The analysis then also provides additional information on consistency.

For this the Jet-Valve has been equipped with an 0.3 mm nozzle and is driven at a low level of 0V, 1V, 1.25V and 1.5V in 4 different spotting grids respectively. Low level voltage of above 1.5V resulted in not producing droplets. Out of each grid 15 spots have then been measured under the optical microscope (Figure (33)). The results can then also be compared to the Jet-Valve camera to find its pixel per mm value.

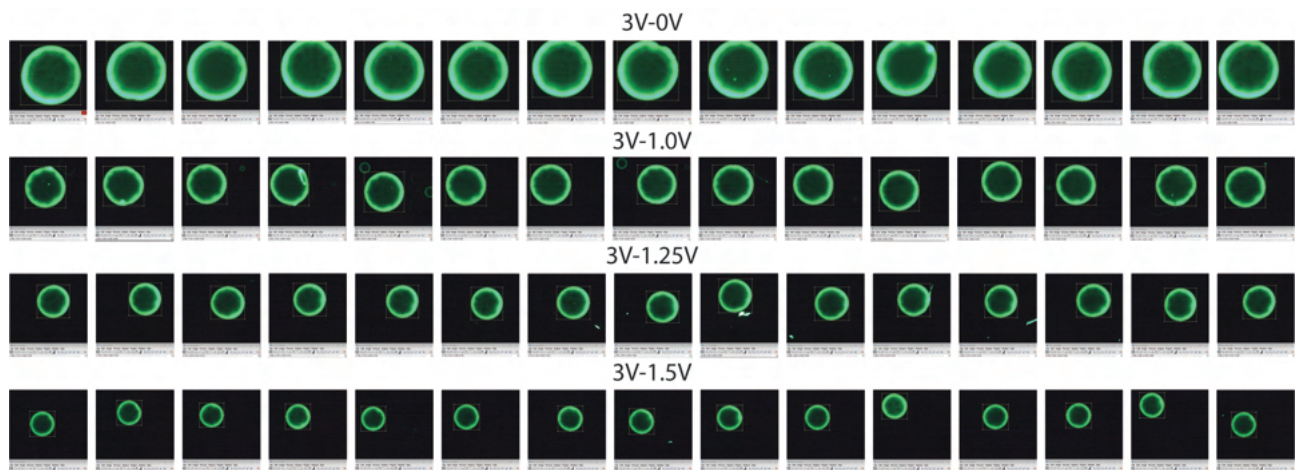


Figure 33: Low level drive voltage experiment

The spots analyzed in Figure (33) were obtained from Figure (36).

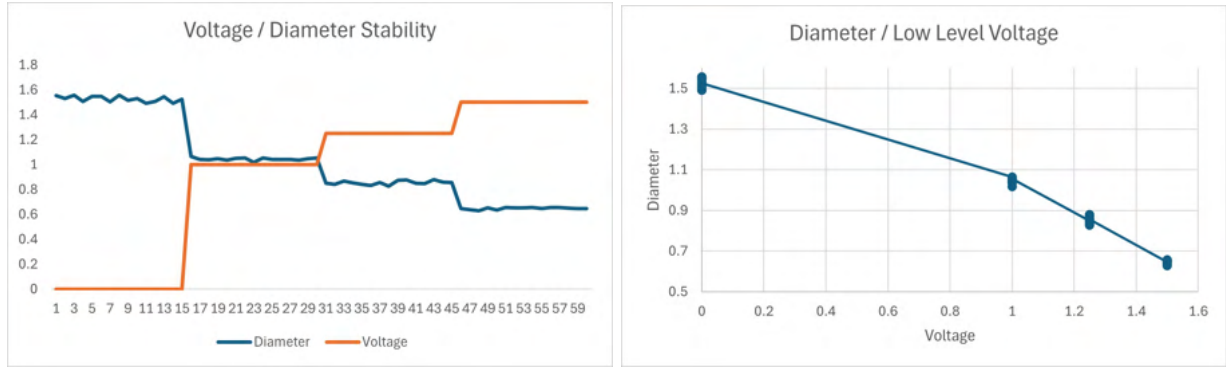


Figure 34: Plot of measurements from figure 33

From which a semi linear relation between the different active driving voltage can be seen in 34.

8 Discussion

Looking back on the objectives of this project (Section (2.2)), major successes were seen and R2R spotting with the designed system was achieved. Still, the performance requirements were not met completely and the partial failures require explanation.

Objective 1

Starting with number one, designing a non-contact dispensing unit, which can spot in a continuous manner and without the need for aspiration. This was achieved, by implementing a Jet-Valve, which can be supplied with working liquid, by means of cartridges, syringes or larger positive pressure reservoirs. For this a dual stage amplification mechanism proposed by SHI et. al [27] was reproduced.

Dependent on the interpretation, the budget criteria can be seen as a success or not. The valve itself, with an approximation for the CNC components given by PCB-Way, 3D printed parts only considered for material cost, an approximated cost for screws and piezo bought VAT free, while sourcing the nozzle from Amazon can be built for 160€. The piezo driver itself comes at 700€ if ordered in Germany, or 466€ if sourced directly from Australia. Because the driver can be estimated to cost less than 90€ if designed in-house, the goal of a unit below 250€ is achievable. Since designing the driver was out of scope for this project and the resulting unit uses a commercial piezo driver, which itself is driven by an additional power supply and function generator, the budget requirement is seen as unsuccessful. Within a full time project, this could have been implemented by designing a driver, that includes:

- Power supply
- Digital-to-analog signal converter
- High voltage amplifier
- Hv-PushPull driver

Despite of achieving the production of a Jet-Valve capable of non-contact dispensing, the design is concluded to not be successful, because the performance requirements have only been partially achieved. The design is capable of continuous spotting at 100Hz theoretically supports driving frequencies up to 249Hz. At that frequency, droplet footprints below 300 microns had been achieved but not been controllable in sizing and positioning. Since Table (5) and Figure (33) show that the achieved minimum repeatable droplet footprints with true Drop-On-Demand behavior had been ≈ 650 microns, and inherently a grid pitch of 600

microns would also not be possible, the performance of the device is not satisfactory. In addition, spotting a grid the size of two A4 paper sheets had also not been achieved, even with the greater droplet footprint.

The reason for the failure of those three points is that the fluid supply path design and consideration was neglected. During the project ideation phase and review of articles, the importance of constant pressure in the fluid path, was not realized. Even though multiple found literatures [27][20][6], which clearly show the dependency of droplet volume and formation on the supply pressure, had been read, the connection was not made, and no pressure control was included in the initial design phase.

That not having pressure control would be a big issue did not surface until hitting the ending phase of the testing and validation period and testing with the actual polymer cocktail. Previously the system had only been tested with mineral oil and the iteration one needle assembly using a custom teflon disc sealing. The combination of those systems brought the realization, that a closed fluid path as described is required for the creation of consistent droplets, as described in Section (5.3), it was not seen, that the disfunction of the teflon disc sealing, was what enable the creation of droplets, even when not applying active back pressure. This worked, because the described vacuum was broken by air passing the sealing, removing the ejection opposing force, allowing droplets to form. Even though applying too much positive pressure by hand, resulted in a leakage phenomena as also witnessed by SHI et. al [9], the project continued without further consideration to this, as the system was capable of spotting grids with consistent positioning and sizing, for example Figure (35).

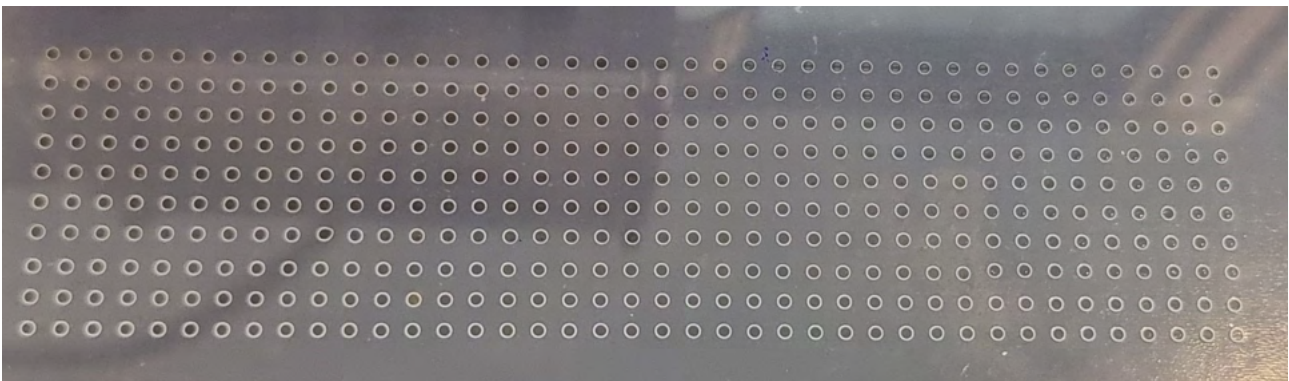


Figure 35: 40 by 10 grid of mineral oil at 3mm spacing

Results like (35) gave the assumption, that the design was working and ready for further testing during the testing and validation.

That this had been a misconception was shown, when first testing with the polymer cocktail. Similar to when testing with mineral oil the leakage phenomena was witnessed. What had previously not been seen, but obvious if given actual consideration, is that upon leaking the working liquid also moves up the needle assembly. In difference to mineral oil, which had acted as a lubricant in that case, the toluene was now evaporating on the needle, only leaving residues of polystyrene. Those residues quickly accumulated, to a thick layer surrounding the needle and then changing its width tolerances resulting in getting stuck inside the assembly. This made recoil of the needle impossible and therefore broke the mechanism.

At the time of this issue arising, the supplier for precision sealing, needles, needle assemblies and nozzles [JHMS Technology](#) had been found, who is likely to be a supplier of those components to all major companies producing the industrial Jet-Valves, as they themselves rate them compatible with Axxon, Nordson EFD, VERMES, Techcon Systems, Graco, GKG, ANDA, Mingseal, GK-PRETECH, Asymtek, Fisnar, Musashi Engineering, Precision Valve & Automation, GPD Global.

Detailed information about the assembly process and sophistication of industrial valves had previously been unavailable to such an extent. Making them very valuable in showcasing the type of special oil sealing, likely to also have been used by SHI et. al [27], the main reference paper to the Jet-Valve designed in this project. Even though the price range of 30-60\$ per component, made them available for consideration, them being a chinese manufacturer made those components out of reach to be implemented in time.

After not finding such precision sealings by local distributors, like RS, a final try was made to create a new custom sealing. Even though not long-term compatible with toluene, a circular disc was stamped out of 3mm thick rubber and modified with a concentric hole. The center hole was expanded with drills from 1.9mm upwards in 0.1mm steps until semi easy sliding through it with a 2mm needle was possible under lubrication. This had been achieved with leather stamp, a 2.5mm drill and 3D printed tool as concentricity guide. This proved to be a success, making the system sealed and able to hold against positive pressure exerted by hand through the syringe.

Now having a sealed system the vacuum condition as described in Section (5.3) was holding, making spotting impossible without positive pressure in the fluid path. Until this point no solution, again due to poor project planning had been adapted. Using a laboratory syringe pump was initially considered, but discarded, due to the assumption that making a custom syringe pump, which could be driven by the same control board as the R2R-Rollers would provide simpler integration into the system. That assumption was then fueled by fast design and time to successful running as part of the system.

Only after testing with the actual single micro steps it was witnessed, that driving a stepper motor in such a way, results in continuous current excitation of the coils and without any valuable movement providing cooling, leaving the motor rising in temperature until completely disabled. The temperatures at these driving conditions rose to from "too hot to touch" to "melting the PETG mounting components", hence approximately 85°C (39). Which a stepper motor as stated in the data sheet can handle, but is of no use in a real life. After this failure there was no time to test any different directions anymore. But possible solutions can be:

- Active cooling of said stepper motor
- Using air pressure cartridges as every other Jet-Valve supplier and researcher in all found reference papers

Because no constant pressure in the fluid path was given during testing, the actual achievable minimum droplet footprint diameter that this system could produce was not found, because no repeatable test conditions could be established. This does not affect the results gathered in Figure (33), as they had been gathered using the syringe pump before failure.

Objective 2 & 6

The objective, that the device should be controlled by an application that users without relevant programming skills can use the dispenser is seen as a success, because the resulting application can be started as an executable without the need of a programming IDE. The functionality can be understood after verbal instruction or after reading an instruction sheet. The application would still benefit from further styling, improvements and a help section inside the application. The live visualization is working, showing the current progress of the spotting process but could benefit from additional parameter visualization, e.g passed time, expected time to finish and a display of the progress in percent.

The steps to download the create G-code file onto an sd-card, insert it into the printer and manually start the process have been completely eliminated.

Objective 3

The creation of a Roll-2-Roll Add-On for the spotting on larger sheets is seen as a success. The De- and Re-Rollers are implemented on a separate control board while the paper feeder is reusing the y-axis stepper signal of the original Ender printer. Switching between R2R and build plate mode is as simple as taking the paper feeder of the build plate and switching its wire back into the original y-axis motor.

Although seen as a success regarding the success criteria (Section (2.2)), the R2R platform implementation could be improved if an appropriate control board supporting up to date marlin firmware versions would have been chosen. Using such a board would make it possible to eliminate the need for modification of the original printer.

Objective 4

Implementing the Jet-Valve dispensing head as an easy to remove add-on is seen as a success since the average mounting and unmounting time is well below 5 seconds.

The fact, that this project removed the original FDM-head from the Ender 3 Pro 3D printer is not seen as contradictory to the "remove need for FDM head removal" requirement because the mounting plate can easily be adjusted to mount to any other 3D printer with its FDM head still mounted.

Hence the Jet-Valve can be used next to FDM heads and removed when not in use.

Objective 5

Adding a multiple solution dispensing head was not realizable within the given time frame. Such a system would require the manufacturing of another Jet-Valve, which would simply take time and resources and the point proven would have small impact on the project's results.

The theoretical implementation of this would require either another piezo driver or a multiplexing circuit, connecting one piezo driver to multiple Jet-Valves. As this was a "nice to have" the implementation would have only been done if all other tasks, had been finished way a head of time, which had not been the case.

9 Conclusion

The project is seen as successful. All three main goals were completely or partially achieved and the knowledge gained in the categories of system design and driving a project from ideation to finish solitarily is seen as invaluable.

In hindsight it can be seen, that the project achieved to build a Roll-2-Roll platform for the creation of optodes. The failure of achieving the planned performance of the device as described in 1 can be explained and further solutions are proposed. It is acknowledged, that more time should have been spend on the theory of jetting and a respective analysis should have been conducted in the beginning of the project. With this in mind the objectives can be summarized by Table (9):

Objective #	Description	Success
1	Design dispensing unit	a) (XX ✓), b)✓, c)✓, d) X
2	Create desktop application	a) ✓
3	Develop Roll-2-Roll Add-On	a) ✓, b)✓
4	Develop Clamp-On mechanism	a) ✓
5	Add multiple solution dispensing	X
6	Enable remote control	✓

The overall objective described as the scalable and generalized deposition of multiple chemicals in various patterns at sub-mm spacing (1.1) can be achieved with the designed system. The problem at hand towards achieving this is the required changing of the liquid, as only a single jetting body is available. This introduces great risk of cross contamination and potential offset in volume, introduced through the required de- and re-mounting of the nozzle during the filling process (Appendix (42)). A robust system requires multiple dispensing heads.

As expected, the project provided a deep dive into the field of mechatronics, requiring the fusion of mechanical, electrical and software systems. Overall the project is seen as successful, even though the performance requirements have not been met.

A continuation of the project would provide tasks like the complete parametrization of the system, implementation of footprint control loops like PID based on the camera vision feedback and general improvement of the design sophistication. Due to the main mechanism only requiring a width of 10 mm, thinning down of the Jet-Valve below 20mm is reasonable and be a step towards, having several mechanisms being mounted in parallel, while being sequentially driven through a multiplexed driver.

10 Appendix

Spherical Cap Volume

With the footprint diameter 300 microns and approximated height of 75 microns ($\frac{1}{2} * radius$) due to contact angle of 60° the volume can be found by:

$$D = 300 \mu\text{m}, \quad h = 75 \mu\text{m}, \quad r = D/2 = 150 \mu\text{m}. \quad (5)$$

The volume of a spherical cap is given by: [18]

$$V_{\text{cap}} = \frac{\pi h}{6} * (3r^2 + h^2). \quad (6)$$

Substituting $D = 300 \times 10^{-6} \text{ m}$ and $h = 75 \times 10^{-6} \text{ m}$ into (7):

$$\begin{aligned} V_{\text{cap}} &= \frac{\pi (75 \times 10^{-6})}{6} (3 (150 \times 10^{-6})^2 + (75 \times 10^{-6})^2) \\ &\approx 2.87 \times 10^{-12} \text{ m}^3 = 2.87 \text{ nL}. \end{aligned} \quad (7)$$

Footprint Diameter D [μm]	Base Radius $r = \frac{D}{2}$ [μm]	Height $h = \frac{r}{2}$ [μm]	Volume V_{cap} [nL]
300	150.0	75.0	2.87
650	325.0	162.5	29.21
800	400.0	200	54.45
1500	750.0	375.0	276.12

Table 6: Estimated droplet volumes for various footprint diameters

Volume to fill 2 A4 pages

Finding the amount of liquid required to fill two A4 pages can be described by:

$$\begin{aligned} N_{\text{page}} &= \left\lfloor \frac{210 \text{ mm}}{0.6 \text{ mm}} \right\rfloor \times \left\lfloor \frac{297 \text{ mm}}{0.6 \text{ mm}} \right\rfloor = 350 \times 495 = 173,250, \\ N_2 &= 2 N_{\text{page}} = 2 \times 173,250 = 346,500, \end{aligned} \quad (8)$$

$$V_{\text{spot}} = 2.9 \text{ nL} = 2.9 \times 10^{-9} \text{ L},$$

$$V_{\text{total}} = N_2 V_{\text{spot}} = 346,500 \times 2.9 \times 10^{-9} \text{ L} = 1.00485 \times 10^{-3} \text{ L} \approx 1.005 \times 10^{-3} \text{ L} = 1.005 \text{ mL}.$$

Dispensing Frequency Calculation

Expecting to work until the ideal print velocity of an [ENDER 3 PRO](#) of $v = 55 \text{ mm/s}$ and desired droplet center spacing of $d = 0.6 \text{ mm}$, the minimum dispensing frequency f is given by:

$$f = \frac{v}{d}. \quad (9)$$

Substituting values:

$$\begin{aligned} f &= \frac{55 \text{ mm/s}}{0.6 \text{ mm}} \\ &= 91.67 \text{ Hz}. \end{aligned} \quad (10)$$

Hence the dispensing unit should be able to spot up to 100 Hz, to have the capability of spotting the required grid, even when the machine is moving at full speed.

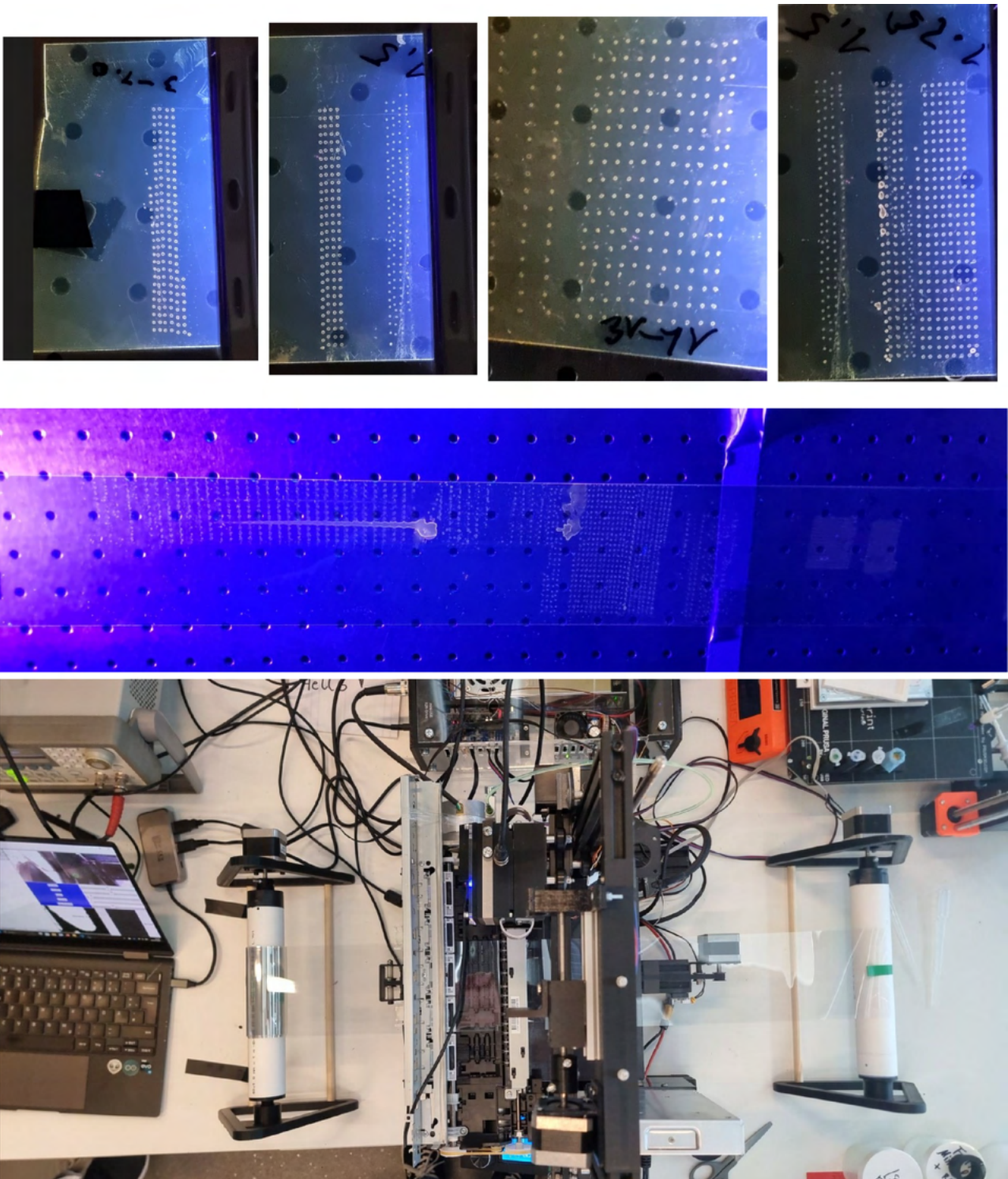


Figure 36: Results Example

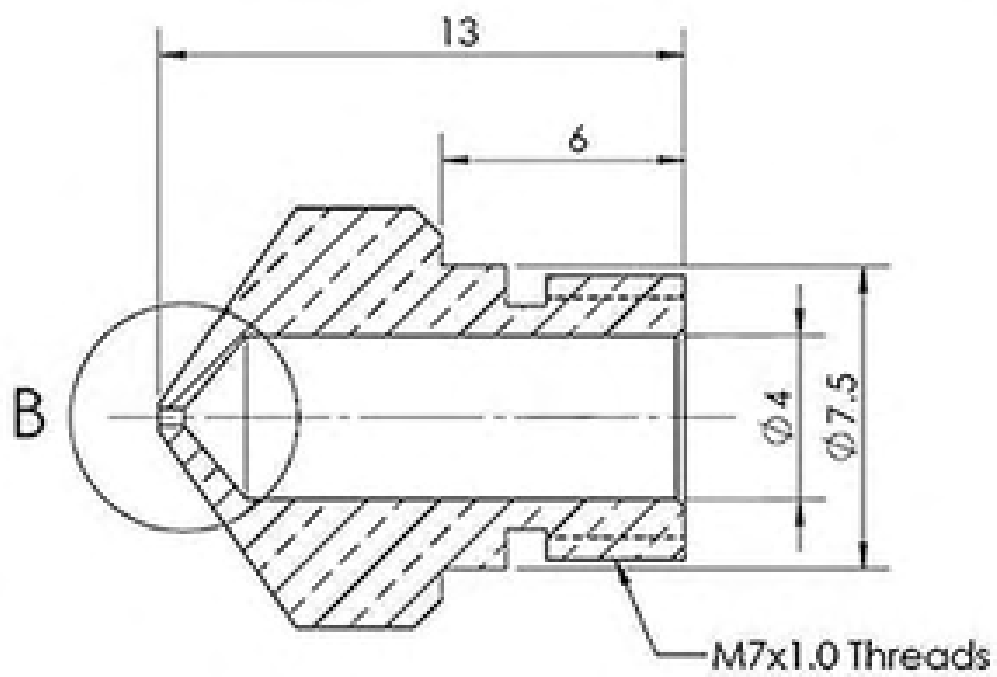


Figure 37: Available drawing for Lufix MK10 nozzles



Figure 38: Expansion of rubber sealing upon contact with toluene

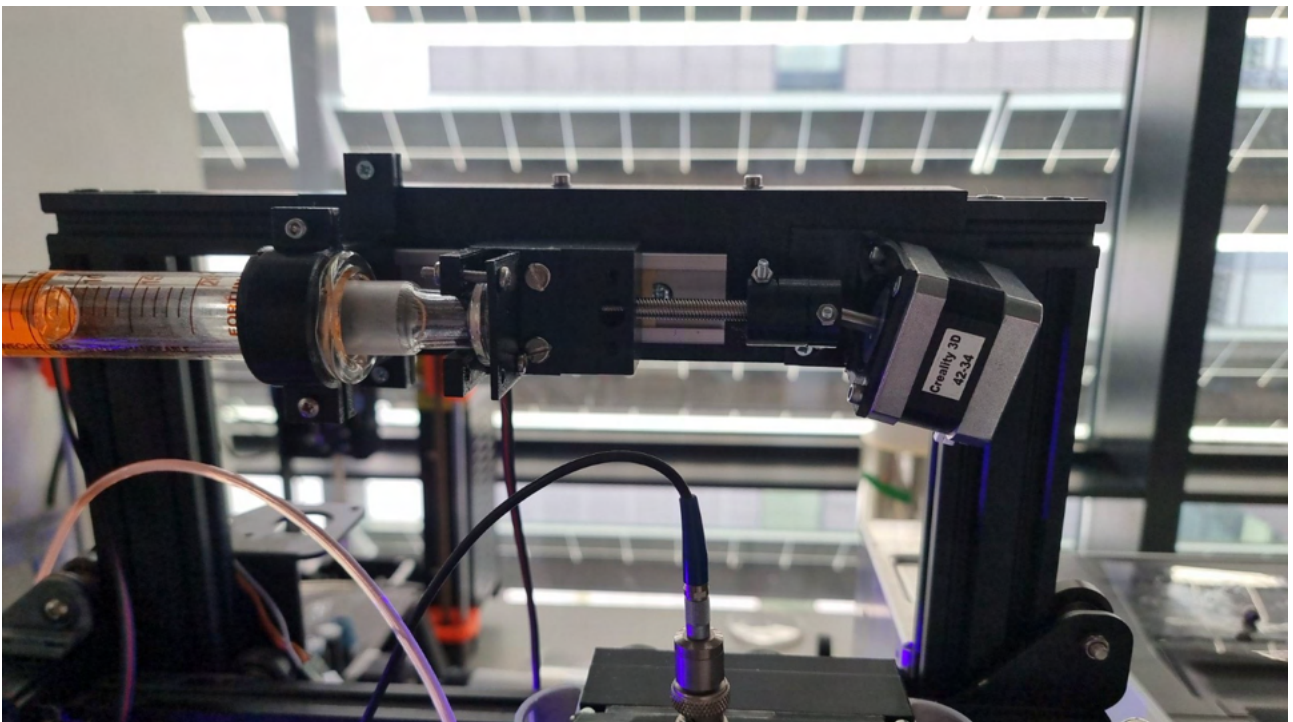


Figure 39: Melting of the stepper motor mount after running at single micro steps

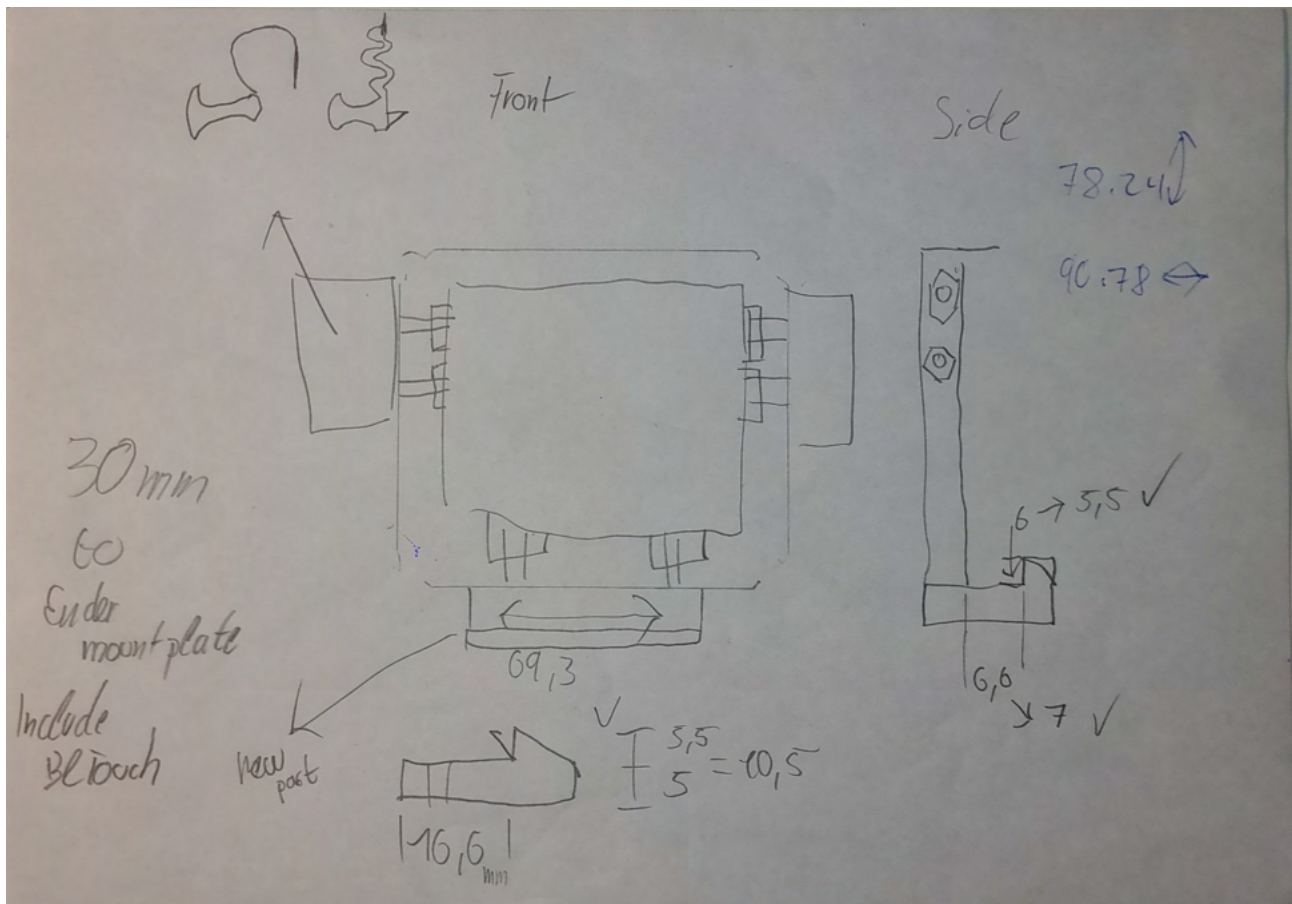


Figure 40: Ideation sketch of the snapping mechanism

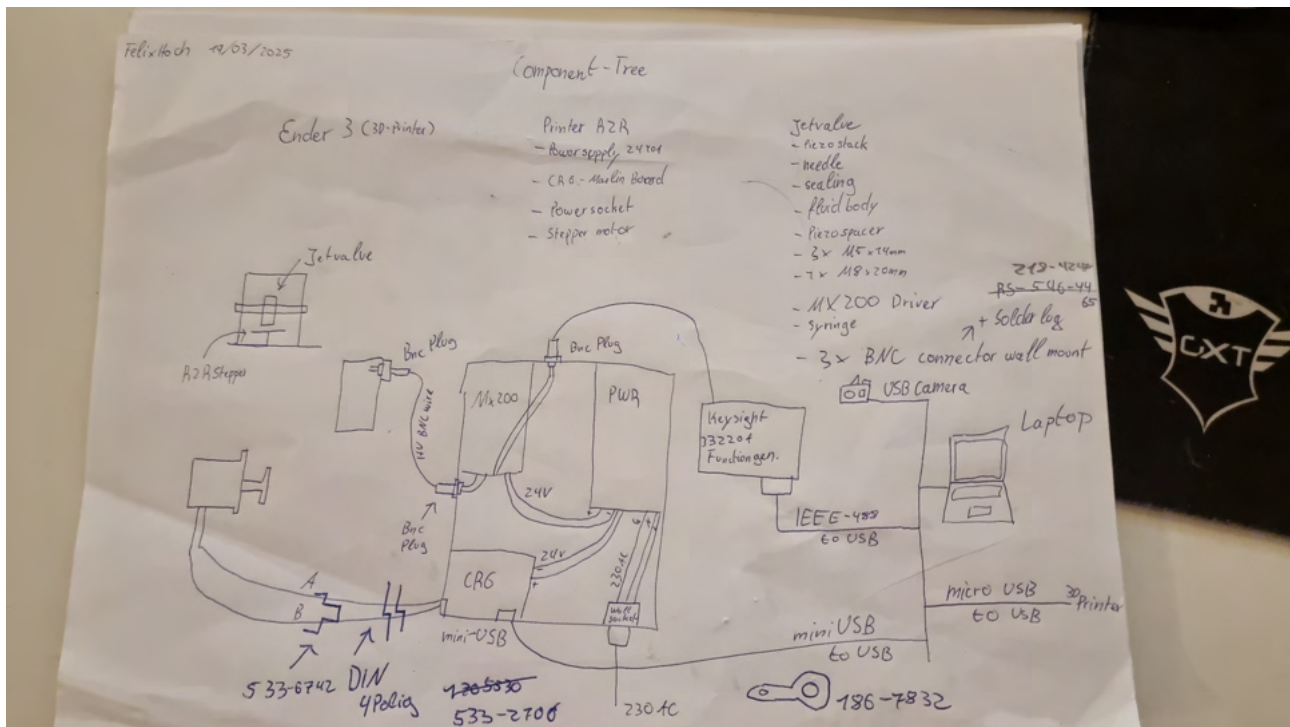


Figure 41: Ideation for ControlBox design

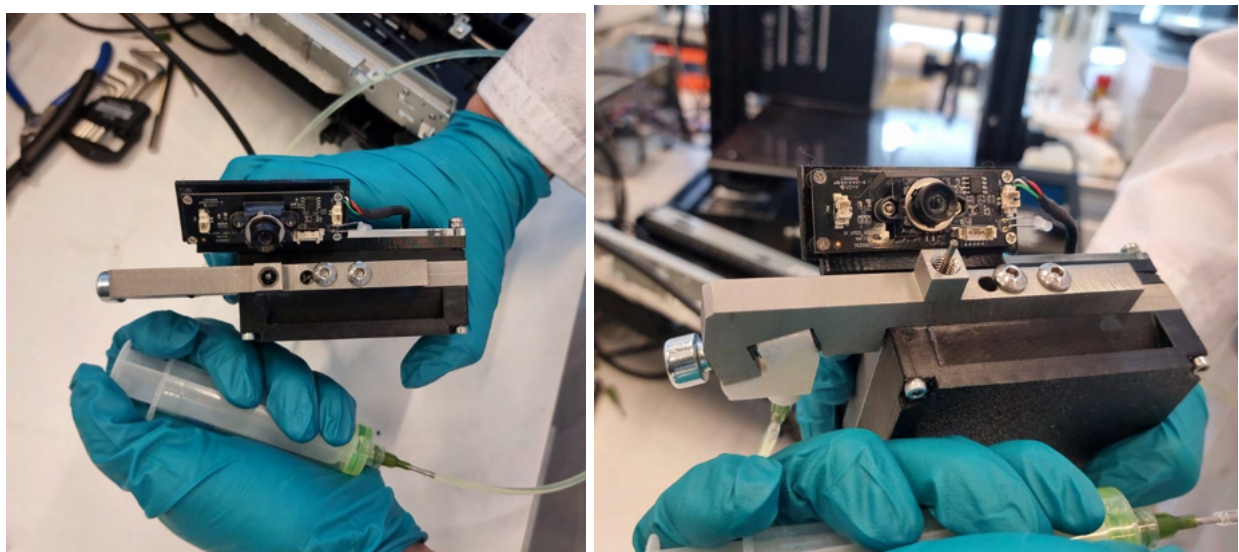


Figure 42: manual degassing of the system through filling with an open jet-valve at the highest point

Figure 43: Technical drawing of the Jet-Valve

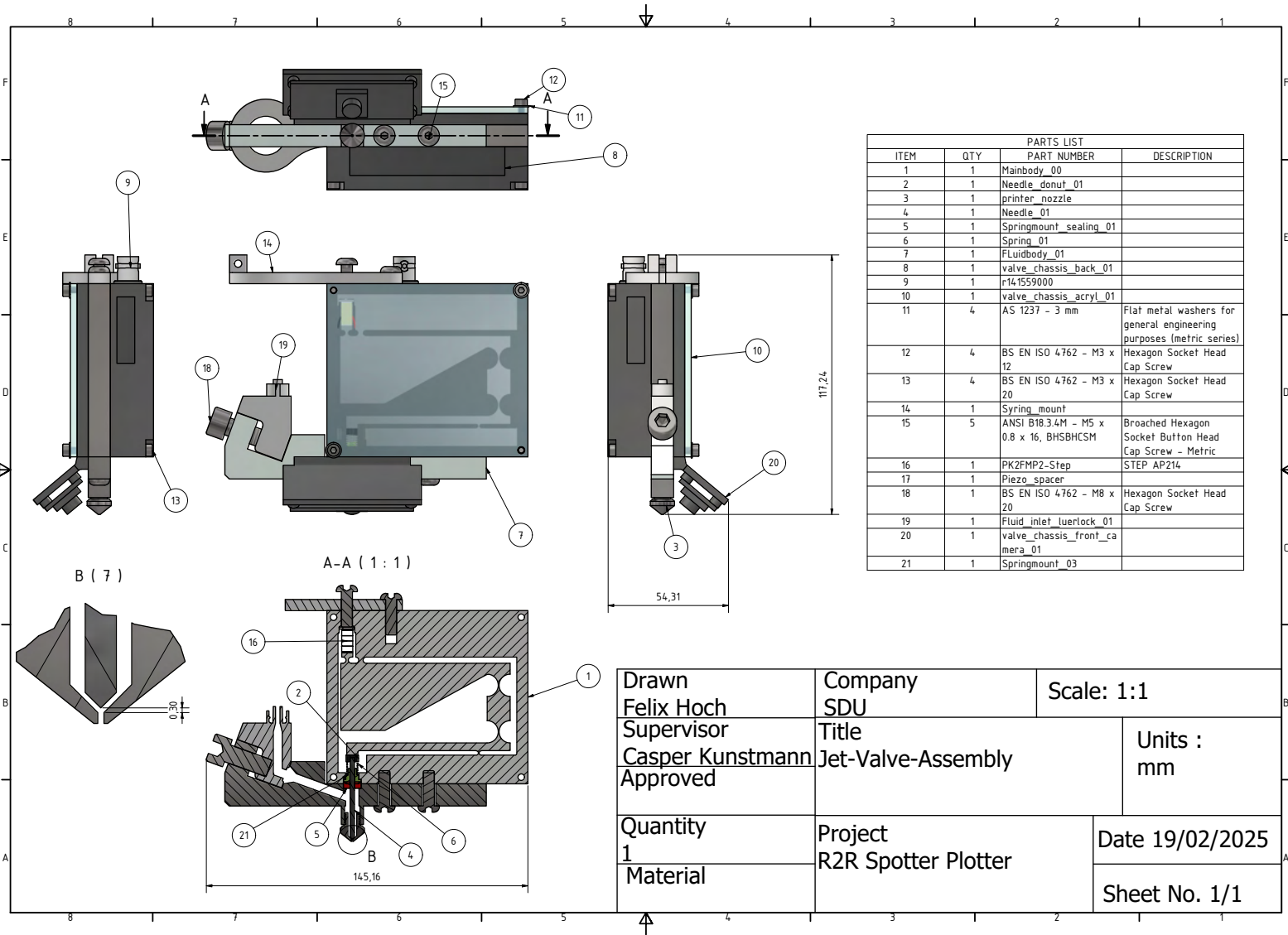


Figure 44: Technical drawing of the Re-De-Roller

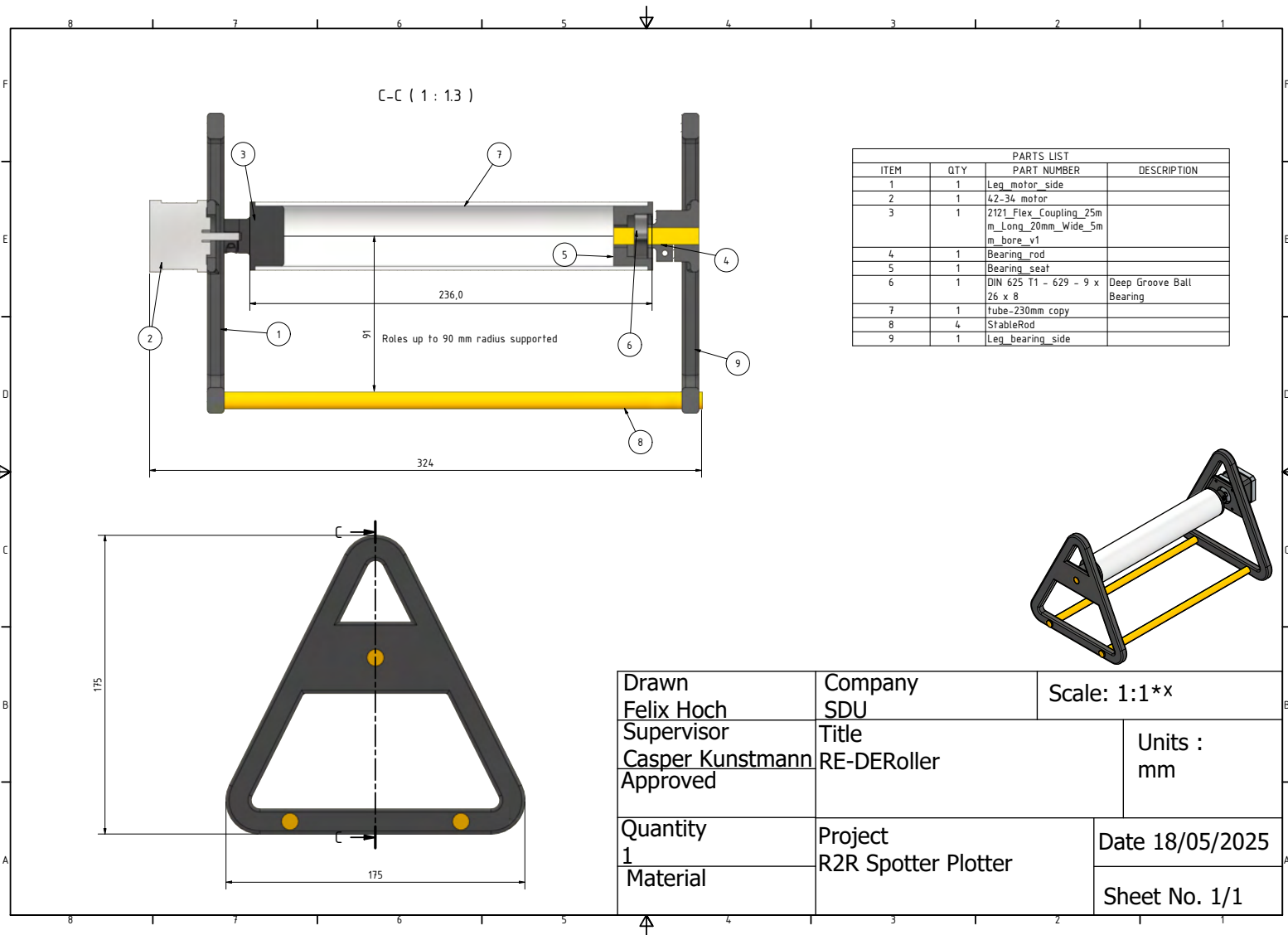


Figure 45: Technical drawing of the ControlBox

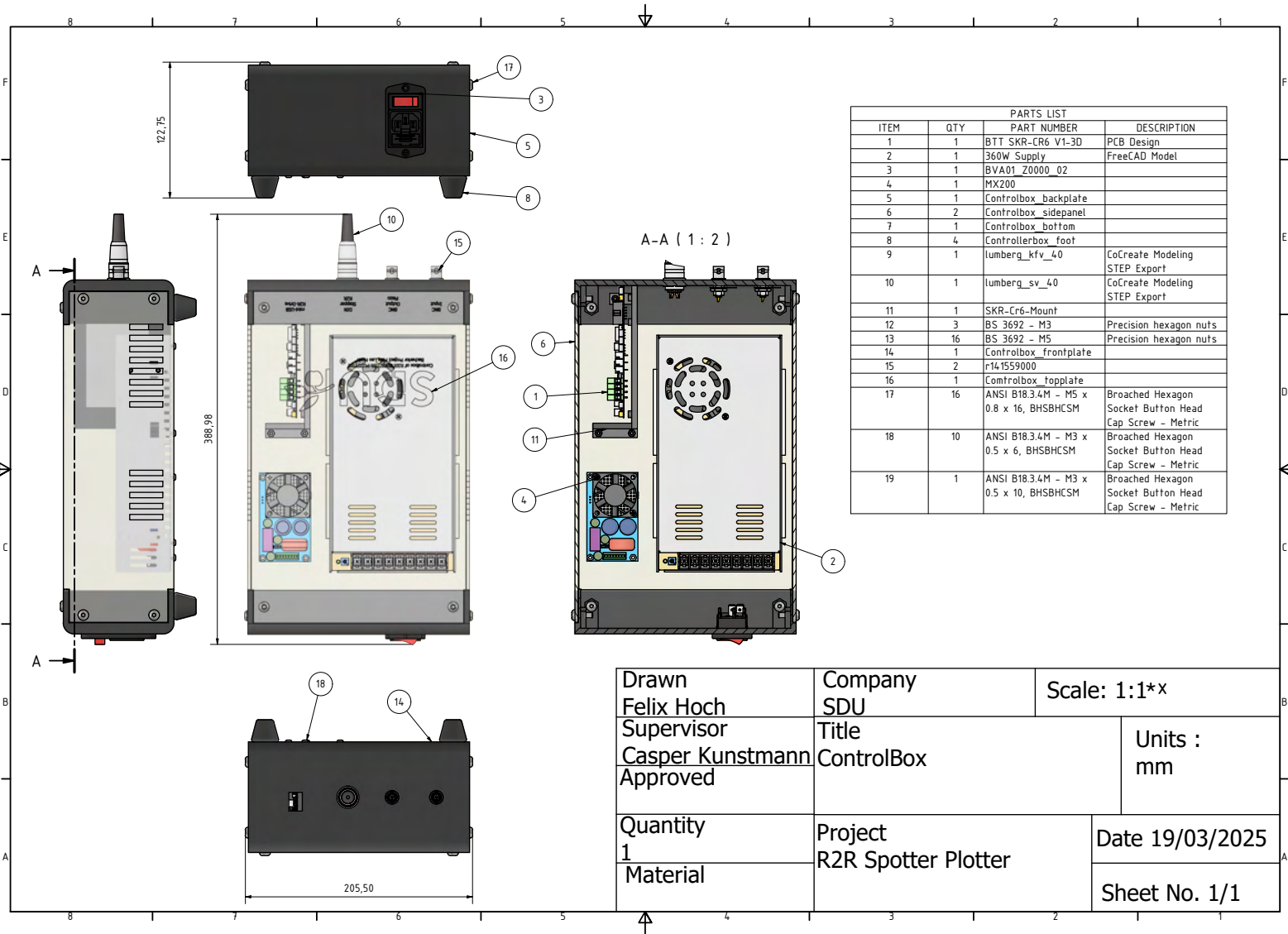


Figure 46: Technical drawing of the Syringe Pump

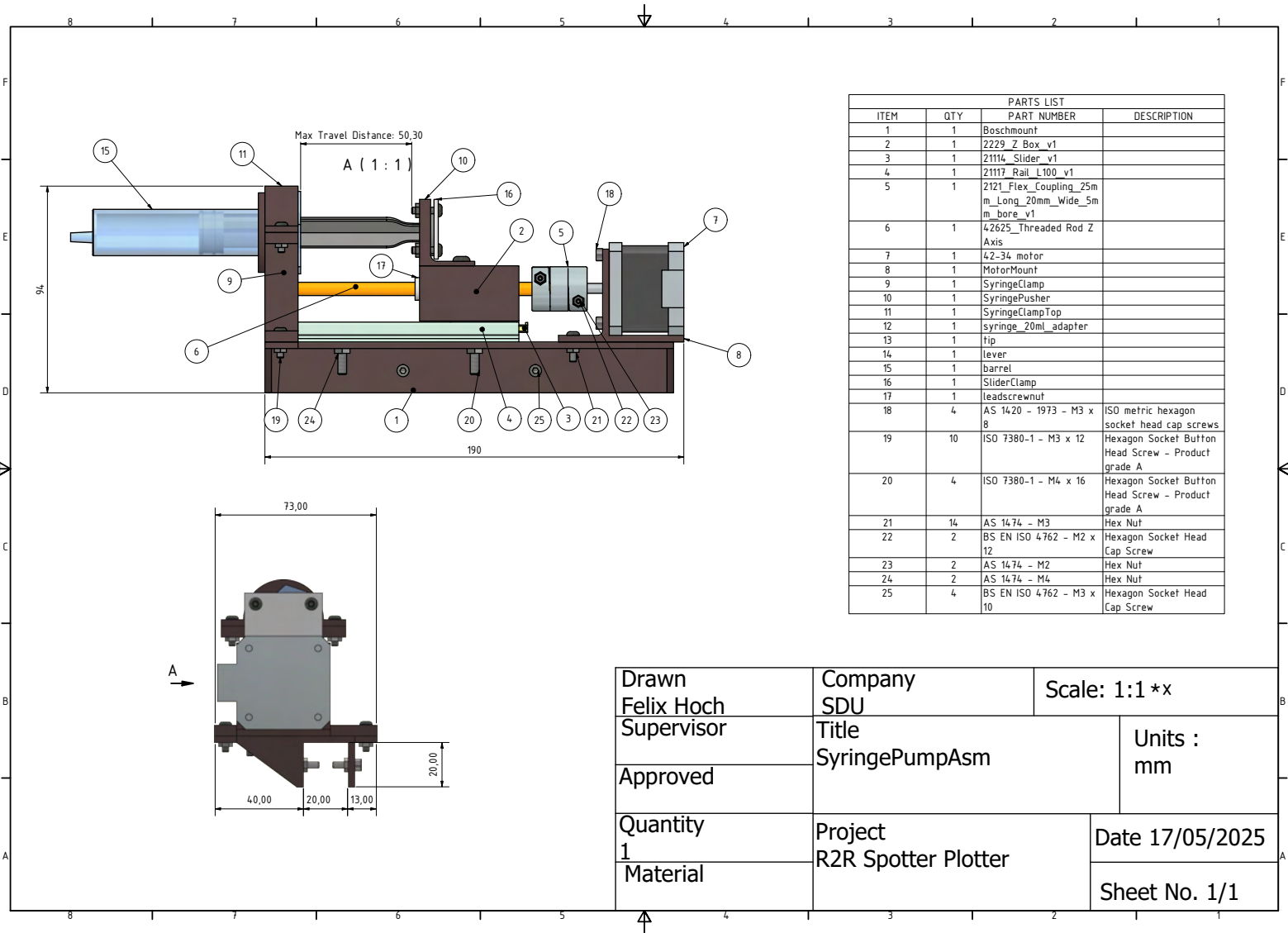
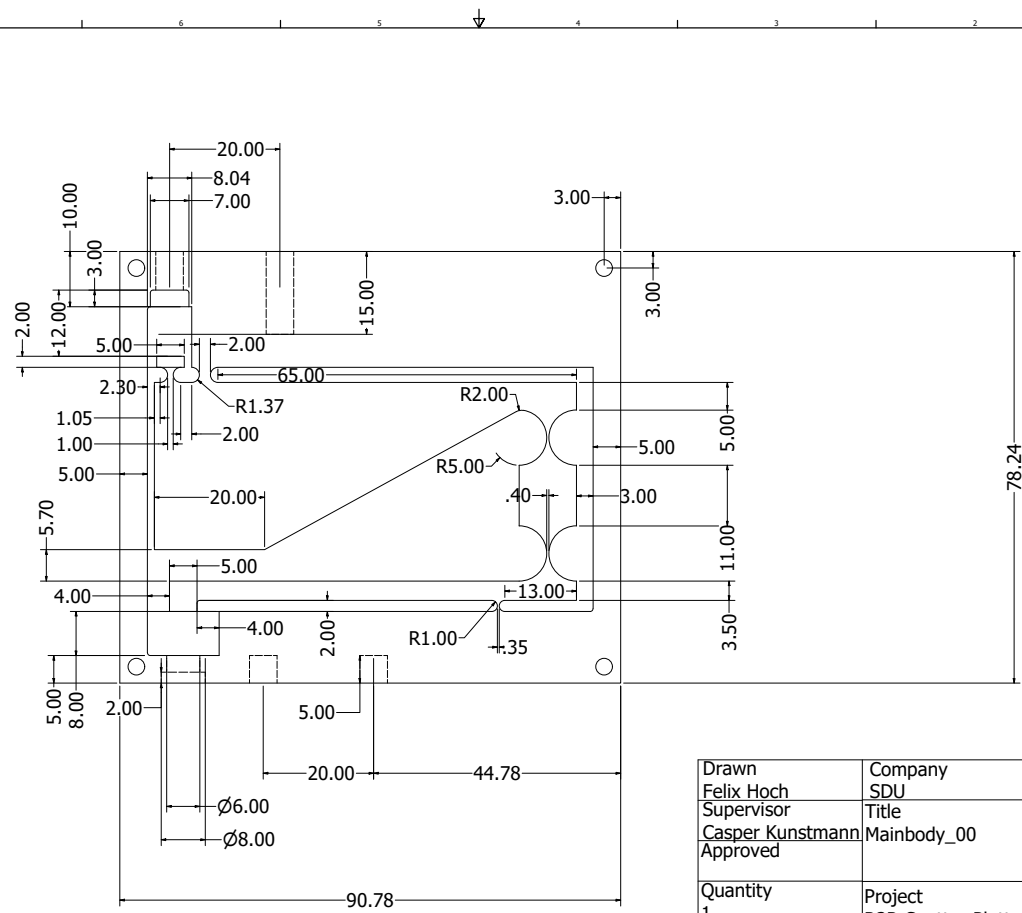


Figure 47: Technical drawing of the Compliant mechanism general dimensions



Drawn Felix Hoch	Company SDU	Scale: 1:1
Supervisor Casper Kunstmann	Title Mainbody_00	Units : mm
Approved		
Quantity 1	Project R2R-Spotter-Plotter	Date 14/02/2025
Material Spring Steel		Sheet No. 1/1

Figure 48: Technical drawing of the Compliant mechanism tolerances

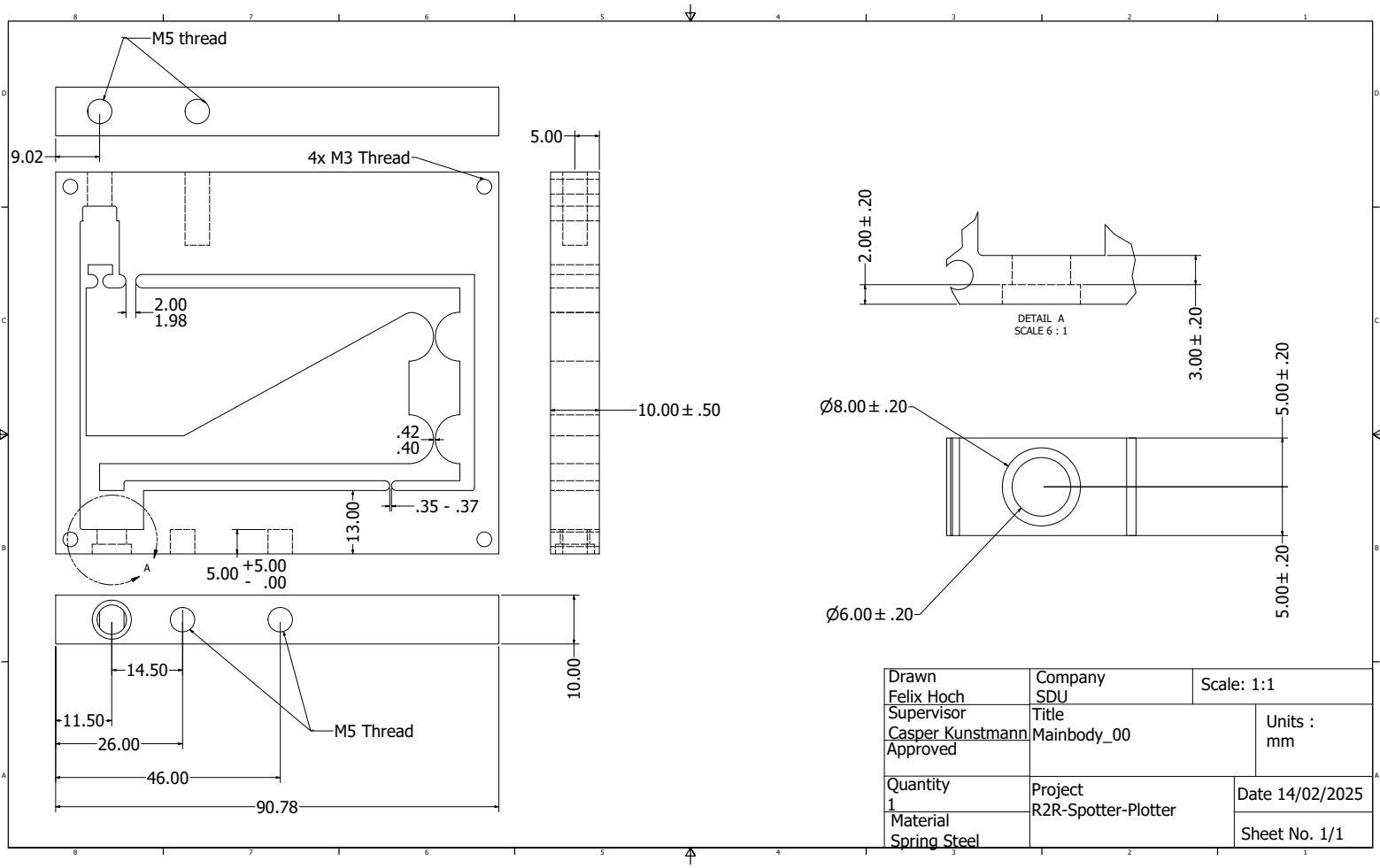


Figure 49: Technical drawing of the Needle

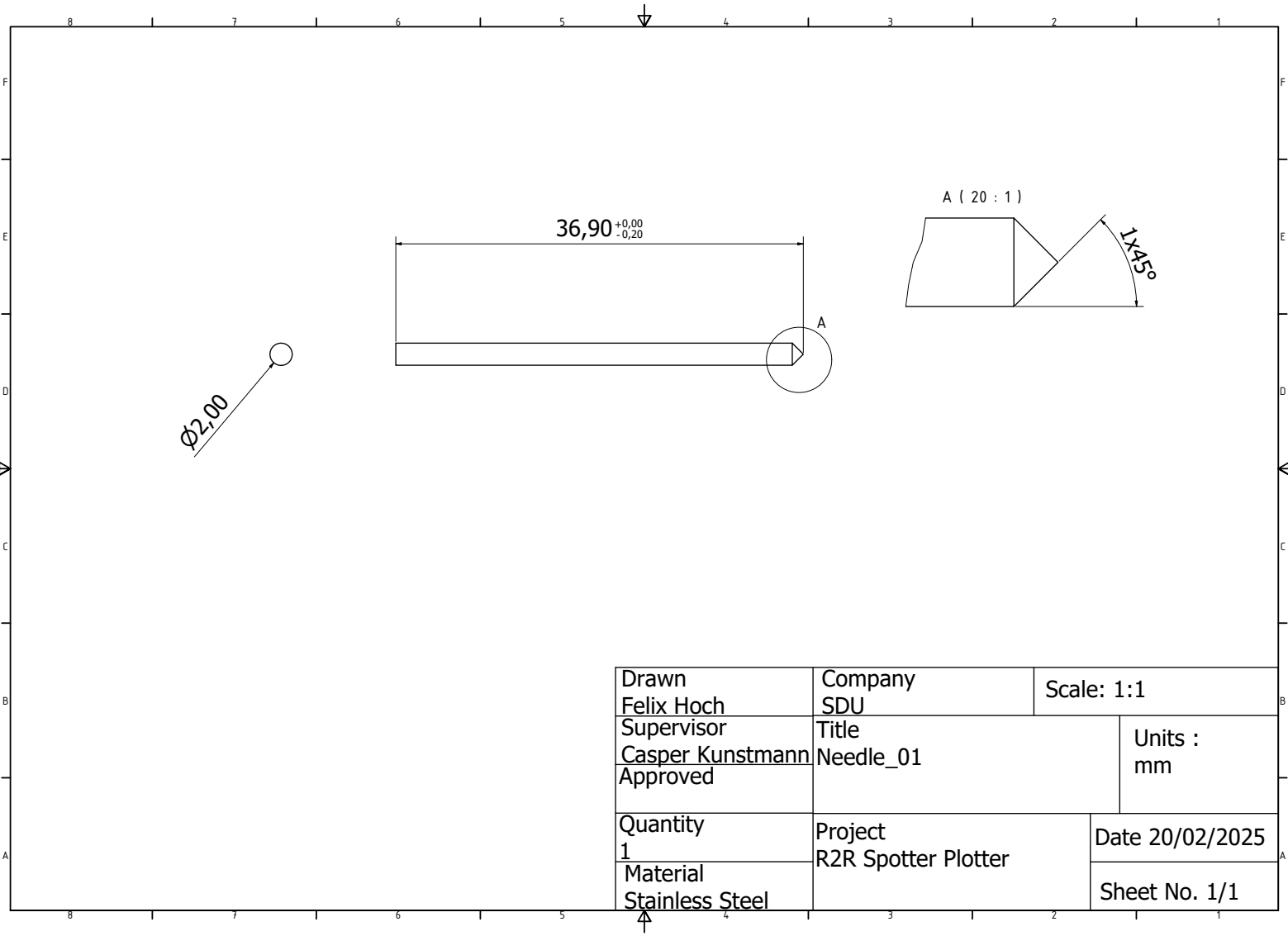
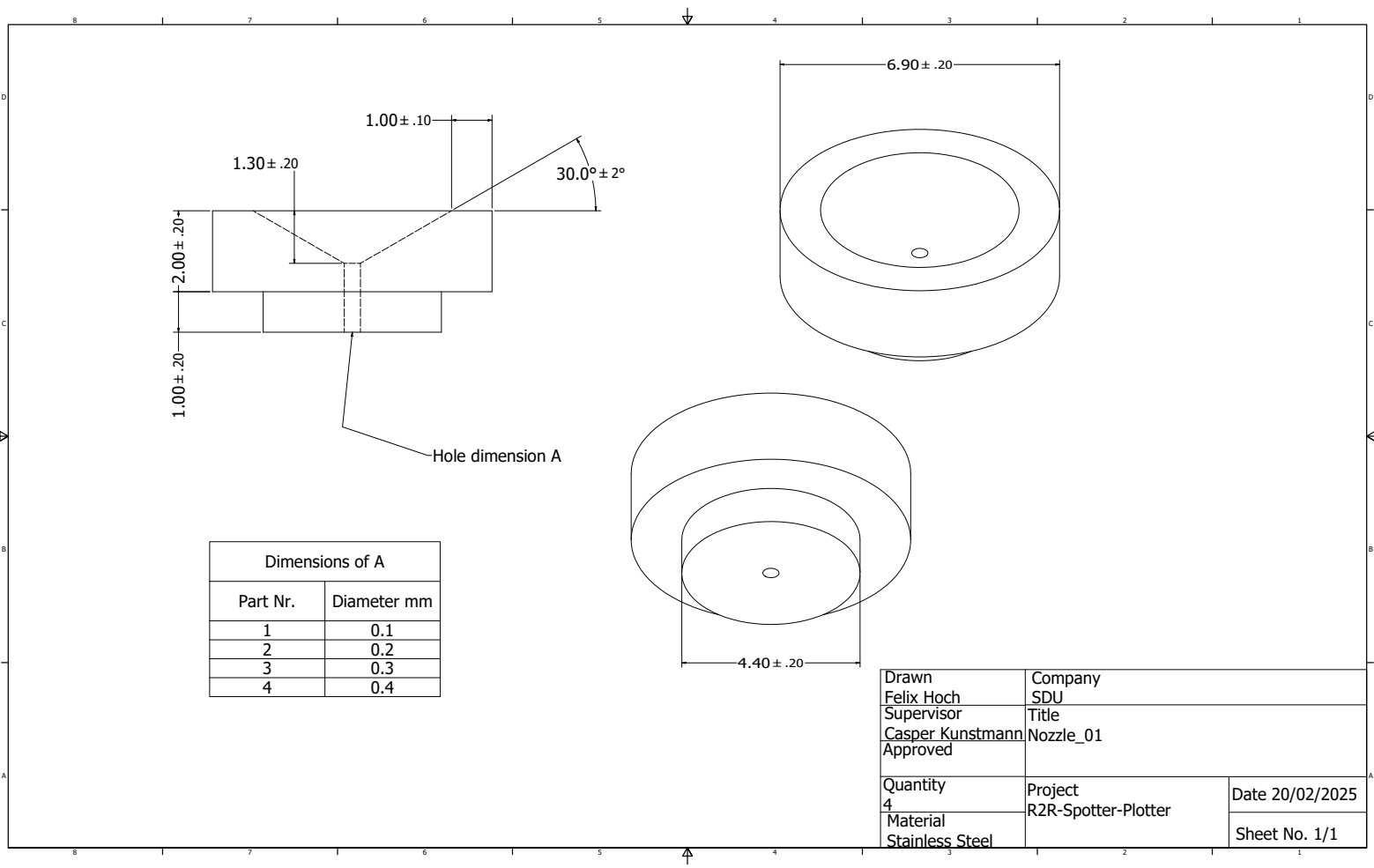


Figure 50: Technical drawing of the Nozzle



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